

Riverdale Public Utility District

Report of Waste Discharge Wastewater Treatment Plant Improvement Project

Fresno County, CA
January 24, 2017



DATE SIGNED 01/24/17

Prepared for:
Riverdale Public Utility District

Prepared by:
Provost & Pritchard Consulting Group
286 W. Cromwell Avenue
Fresno, CA 93711

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Abbreviations

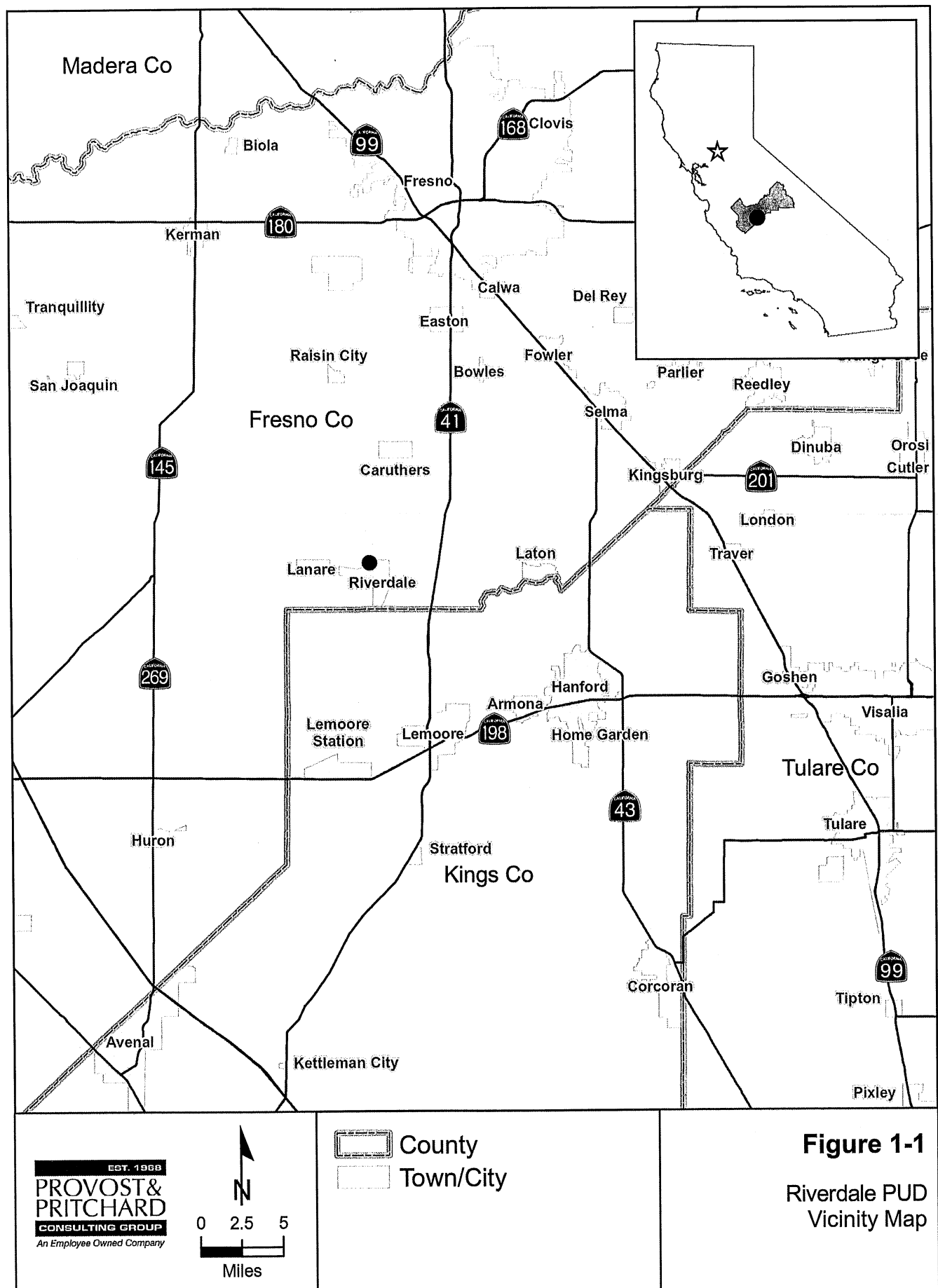
APN	Assessor Parcel Number
BOD.....	Biochemical Oxygen Demand
CEQA	California Environmental Quality Act
CWSRF.....	Clean Water State Revolving Fund
DDW	Department of Drinking Water
DO	Dissolved Oxygen
DWR.....	Department of Water Resources
EC.....	Electrical Conductivity
EDU.....	Equivalent Dwelling Unit
FEMA.....	Federal Emergency Management Agency
FIMP.....	Flood Insurance Rate Map
fps.....	feet per second
gpd.....	gallons per day
gpm.....	gallons per minute
ITRC	Irrigation Training & Research Center
MBR.....	Membrane Bioreactor
MGD.....	Million Gallons per Day
mg/L.....	milligrams per liter
NRCS	Natural Resources Conservation Service
RPUD	Riverdale Public Utility District
RWQCB	Regional Water Quality Control Board
SCADA.....	Supervisory Control and Data Acquisition
SWRCB.....	State Water Resources Control Board
TDS.....	Total Dissolved Solids
TSS	Total Suspended Solids
VFD	Variable Frequency Drive
WDR.....	Waste Discharge Requirements
WWTP.....	Wastewater Treatment Plant

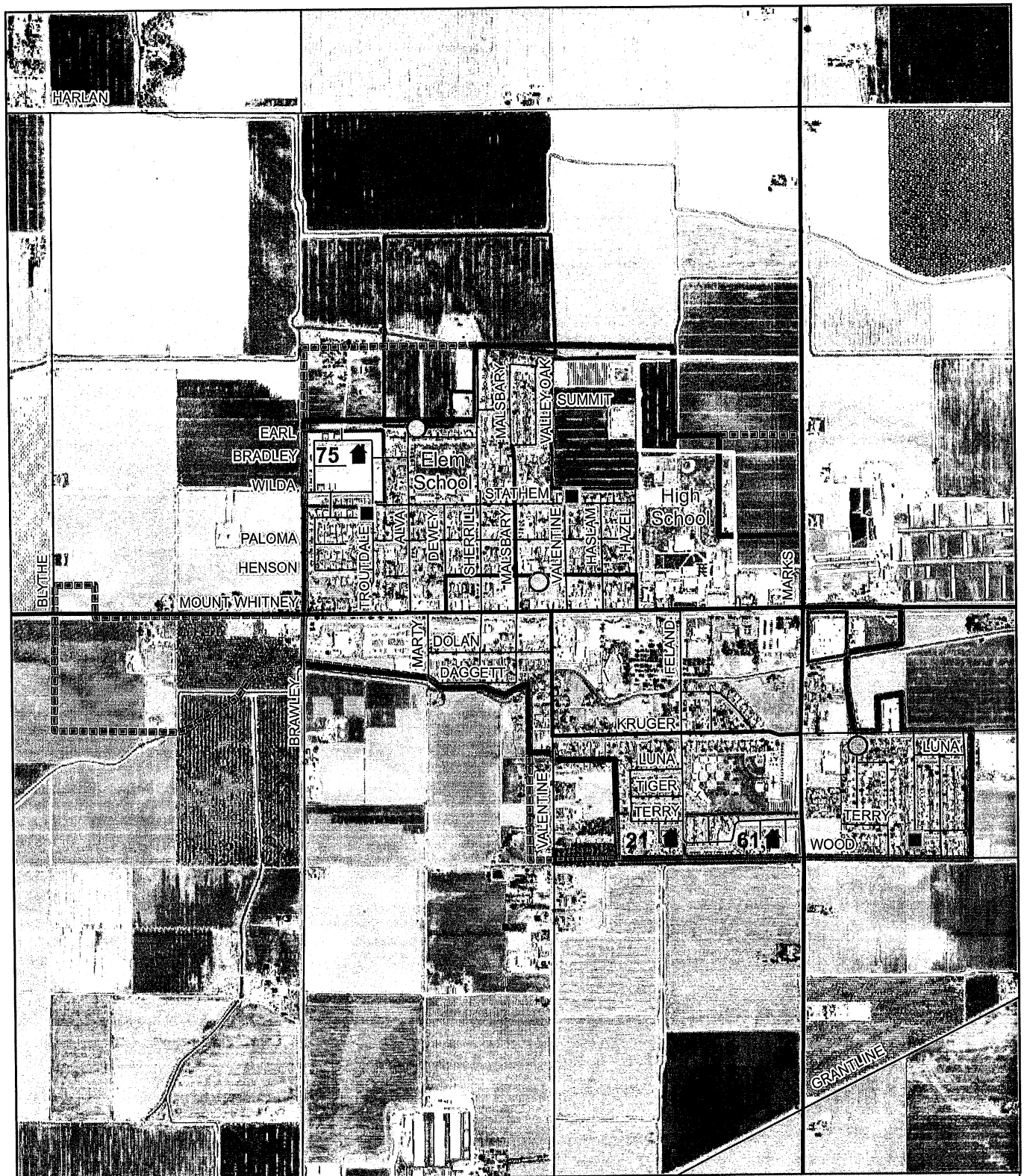
1 General Information

The Riverdale PUD Wastewater Treatment Plant (WWTP) currently operates under a permit issued by the Central Valley Regional Water Quality Control Board (RWQCB), Waste Discharge Requirements (WDR) Order No. 85-252. The permitted capacity of the existing pond-based WWTP is 0.25 million gallons per day (MGD). The facility presently operates at approximately 0.21 MGD (average daily flow), or 88% of the treatment capacity. The RWQCB typically recommends planning for capacity expansion to occur when a system reaches 80% of the design capacity.

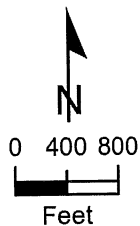
The Riverdale PUD is located approximately 25 miles south of the City of Fresno, in rural Fresno County, California. A vicinity map is included as Figure 1-1. Riverdale is a predominantly agricultural community with no large industrial businesses present. The community had a population of 3,153 in the year 2010, according to US Census data. The current population is estimated to be 3,197 based on the current number of residential sewer service connections (864) and approximately 3.7 persons per household. Figure 1-2 shows a map of the service area.

The WWTP is located east of the Brawley Avenue alignment and south of Cerini Avenue, approximately 1.5 miles north of the Riverdale community. The existing WWTP site is on APN 053-090-011, located in Section 13, Township 17 South, Range 19 East, MDB&M, at Latitude 36° 27' 25" North, Longitude 119° 52' 00" West. An additional 40-acre property was acquired by the District, which adjoins the west boundary of the existing property and consists of the Southwest ¼ of the Northwest ¼ of Section 13. The Van Ness Slough remnant is located in the southeast corner of this additional 40-acre property. The Van Ness Slough remnant will be avoided, limiting the 40-acre site to approximately 33 acres of useable land.





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- | | |
|--------------|------------------------|
| ○ Wells | □ School |
| ■ Lift pumps | □ New development |
| — Minor Road | □ PUD Boundary |
| — Major Road | □ SOI Boundary |
| | □ Proposed development |

Figure 1-2
 Riverdale PUD
 Service Area

2 Wastewater Flows and Loads

2.1 Sources of Wastewater

The land use around Riverdale is primarily agricultural. The large majority of the land use within the RPUD service area is residential, with some commercial, institutional and limited industrial. Figure 2-1 shows the land uses in and around the RPUD service area.

The RPUD currently serves approximately 923 sewer services, including 864 residential services, 47 commercial businesses, 9 churches and libraries, and 3 schools. There are no significant industrial users served by the RPUD WWTP.

2.2 Design Capacity

2.2.1 Projected Growth

Census population data for the years 2000 and 2010 indicate the population increased from 2,416 to 3,153. The population growth rate for this period was therefore approximately 2.7% per year. By contrast, a review of the District's residential sewer connection list in 2005 and 2016 showed that the residential service connections increased from 809 to 864 during that period. The growth rate for this period was less than 1% per year.

Based on the State of California, Department of Finance, Report P-1 (Total Population): State and County Population Projections, 2010-2060 (December 2014), the estimated average growth rate for Fresno County over the next 20 years is expected to be approximately 1.5%.

Based on historic growth rates for Riverdale, and understanding that growth rates will fluctuate over any 20-year period, it is recommended that a growth rate of 1.5% to 2% per year be used for projecting future growth. At this rate of growth rate, and based on the current 923 total service connections, the projected number of sewer service connections in 2035 would be approximately 1,243 to 1,372, or an increase of 320 to 449 connections. This would equate to a population of about 4,600 to 5,080.

The anticipated actual project development in Riverdale should also be considered. Figure 2-2 shows the locations of the anticipated development projects within the District, and Table 2-1, on the following page, shows the current will-serve commitments. It is reasonable to assume that this proposed development will occur within the 20-year planning period. The proposed development would add approximately 383 homes, which is within the 20-year growth range discussed above. This suggests that the estimated growth rate for the District is realistic for facility planning purposes.

Section Two: Wastewater Flows and Loads
WWTP Improvement Project

Table 2-1 Riverdale PUD Will-Serve Commitments

Development Project/Demand	Demand		Remaining Capacity		Notes
	(EDU) ⁽²⁾	(gpd)	(gpd)	(EDU) ⁽²⁾	
Existing System Capacity			250,000		
<i>Capacity used</i>	<i>923⁽³⁾</i>	<i>220,000⁽³⁾</i>	<i>30,000</i>	<i>136</i>	Assumed EDUs
WWTP Expansion (0.325 mgd)	341	75,000	105,000	477	
Jonathan Homes - Fox Hollow (Proposed Units)	26	5,720	99,280	451	Previously Countryside Estates Phase 2. 35 of 61 lots have been built. 26 remain vacant.
CalStar/NASKAT (Proposed Units)	57	12,540	86,740	394	18 of 75 lots have been built. 57 remain vacant.
Malsbary Estate (<i>Phase 1</i>) (Proposed Units)	69	15,180	71,560	325	Inside District SOI
Malsbary Estate (<i>Future Phases</i>) (Proposed Units)	231	50,820	20,740	94	Adjacent to District SOI
Total	1,306	304,260	20,740	94	

Assumptions:

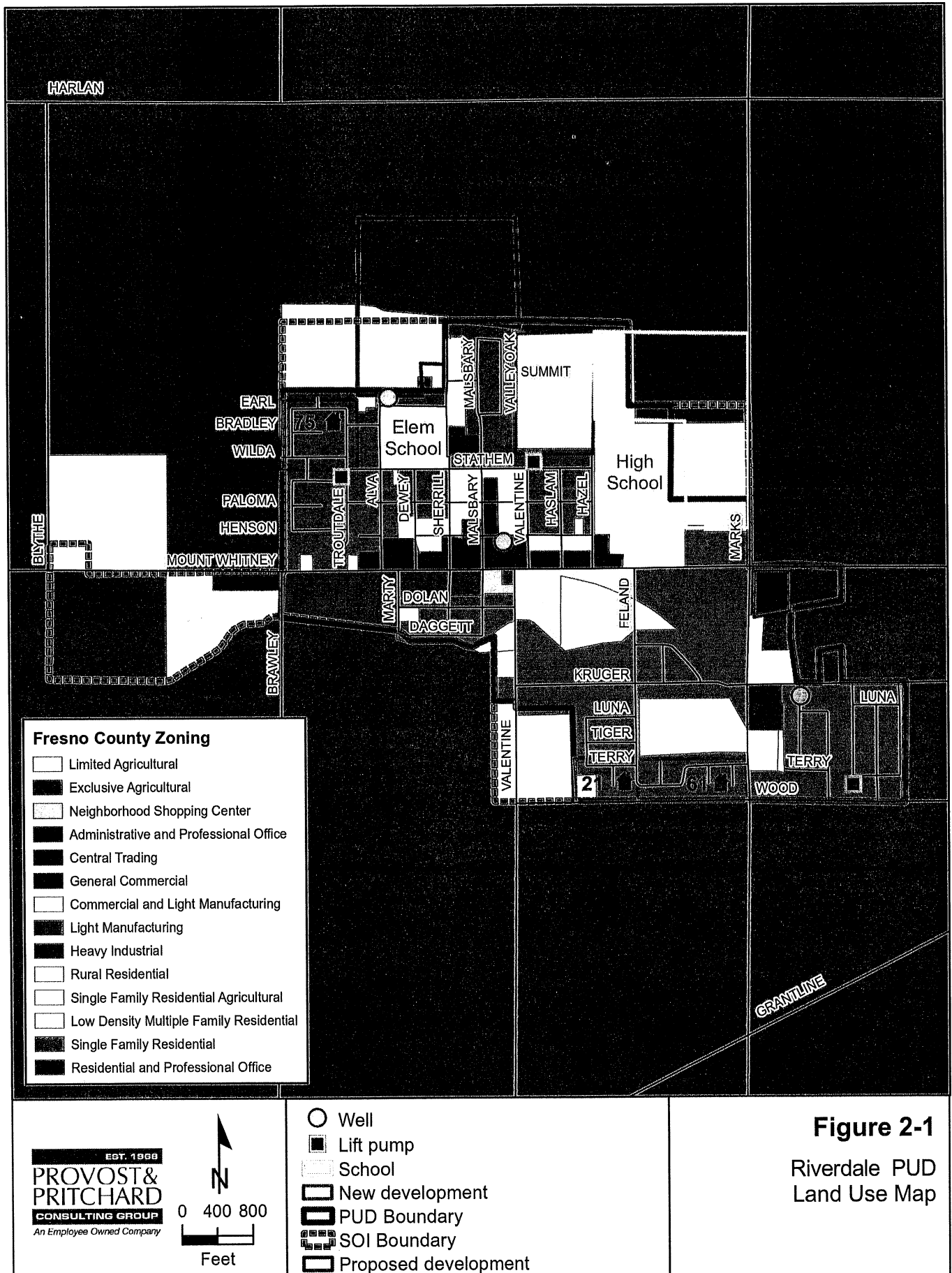
Persons per unit

(Census 2010 Riverdale CDP): 3.7

Sewer load for new development⁽²⁾: 60 gpcd 220 gpd/EDU

Notes:

- 1) Existing capacity used is based on WWTP flow data.
- 2) Capacity projections for future growth are based on recommended design values for new development.
- 3) 2016 Residential Customers: 864 sewer; Total Customers: 923 sewer.



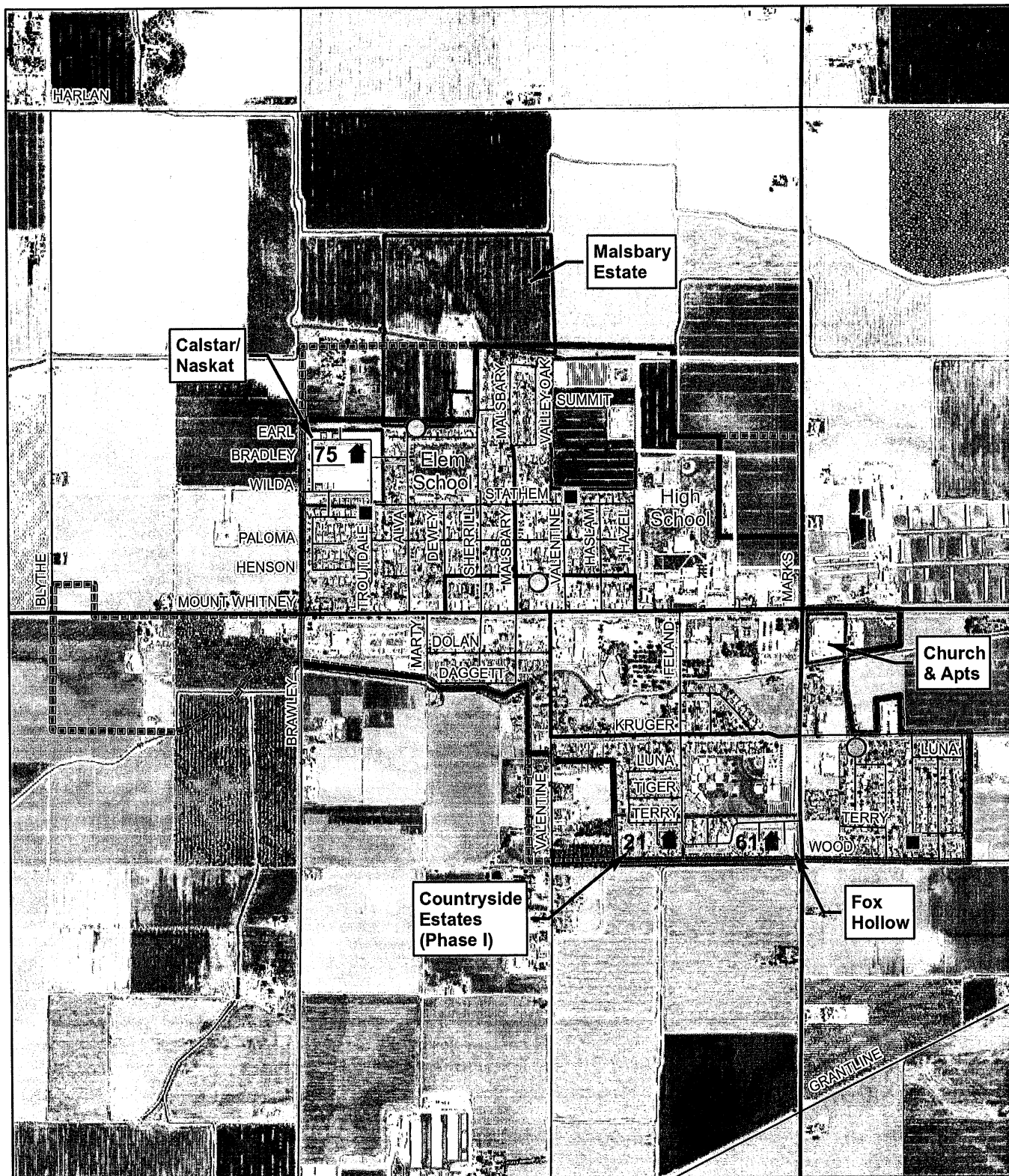


Figure 2-2
Proposed Development

2.2.2 Projected Flows

The WWTP is currently operating near capacity. Typically, the Regional Water Quality Control Board requires planning for capacity expansion to occur at 80% of the design capacity. With current flows nearing 220,000 gpd, the plant is operating at about 88% of its existing design capacity. A typical planning period is 20 years, or the payback period for a loan (if required to fund the expansion), whichever is greater.

Table 2-2 presents estimates of future WWTP flows using the current flow rate observed and projected flow rates in 20 years based on a 1.5% and 2.0% growth rates.

Table 2-2 Projected Future Flows

	Number of Connections	Flow Rate (gpd)	Number of Connections	Flow Rate (gpd)
Planned Growth Rate	1.5%		2.0%	
Current number of connections	923	220,000	923	220,000
Projected increase in 20 years	320	70,400	449	98,800
Total	1,243	290,400	1,372	318,800
Planned design capacity	325,000			

Note: Flow rate calculation is based on a 220 gpd per each new sewer connection (60 gpcd x 3.7 persons per household).

It is noted that, while current per capita sewer flows average about 66 gallons per capita per day (gpcd), per capita flows are expected to decline due to 2013 Building Code requirements for water efficient appliances and fixtures in new homes, which are intended to result in a 20-percent reduction in interior water use. A 2015 study by ConSol, funded by the California Homebuilding Foundation¹, found that indoor water use decreased over 50 percent between 1975 and 2015, with the current use being less than 130 gallons per household per day, or approximately 32 gallons per capita per day at a household formation rate of 4.0 persons per household. That research places water use at a dramatically lower rate than the industry has used until recent years, which has typically been about 75 gpcd. While the study's use rate may be achieved and proved out by future water use measurements, a more conservative number should be used for planning until that aggressive number is proven by experience. The stated goal of the 2013 CalGreen code is to achieve a 20 percent reduction in indoor water use. Applying that percentage to the older 75 gpcd use rate results in a unit flow of 60 gpcd, which has been used to project flows from new development for this report.

It is recommended that the WWTP expansion provide a capacity of 325,000 gpd. The selected expansion capacity will be used to design the WWTP improvements and for the new Waste Discharge Permit flow limit with the Regional Water Quality Control Board.

¹ "Codes and Standards Research Report: California's Residential Indoor Water Use," 2nd Edition, Revised May 8, 2015, ConSol, Stockton, CA

2.3 WWTP Influent Flows

In 2004, the actual WWTP annual average flow rate was 232,000 gpd and the maximum month average was 256,000 gpd. Recent flow data was reviewed for the period between January 2011 and December 2016. The highest annual average flow rate occurred in 2012 at 251,000 gpd, with the maximum month average of 258,000 gpd occurring in February 2012 (see Table 2-3). However, since 2012 there has been a steady decline in flow. Since population has not decreased, the decline in flows is considered to be attributed to water use reductions within the existing households, which is likely the result of the drought and water conservation efforts. It is expected that these declines are not temporary, but that due to ongoing conservation efforts mandated by the State, per capita water use in the existing housing stock will remain at or near this lower level. Assuming that there may be some rebound in per capita use over time, it is recommended that the 2015 maximum month average flow rate of 220,000 gpd be used as the baseline current demand for projecting future flows.

Table 2-3 Riverdale PUD WWTP Average Monthly Influent Flows (in MGD)

Month	2011	2012	2013	2014	2015	2016
January	0.240	0.252	0.245	0.215	0.204	0.208
February	0.235	0.258	0.238	0.214	0.215	0.207
March	0.236	0.250	0.233	0.213	0.213	0.207
April	0.237	0.253	0.227	0.211	0.209	0.202
May	0.233	0.254	0.226	0.207	0.203	0.207
June	0.246	0.243	0.228	0.210	0.204	0.199
July	0.243	0.248	0.227	0.215	0.216	0.188
August	0.246	0.253	0.235	0.221	0.214	0.197
September	0.246	0.253	0.242	0.222	0.215	0.191
October	0.250	0.255	0.241	0.226	0.220	0.198
November	0.252	0.256	0.230	0.222	0.220	0.201
December	0.249	0.241	0.211	0.218	0.209	0.199
Annual Average	0.243	0.251	0.232	0.216	0.212	0.200
Maximum Month	0.252	0.258	0.245	0.226	0.220	0.208

Note:

1. Maximum month shown above represents the maximum month in the year and not the 30-day average daily “dry” weather flow. Sizing based upon the maximum month provides a degree of conservatism to the flow projections.

2.4 WWTP Influent Characteristics

Influent flows to the WWTP are primarily from residential uses, with some commercial and institutional connections. Influent flow characteristics are summarized in Table 2-4. The sampling results for influent Biochemical Oxygen Demand (BOD) and total suspended solids (TSS) are very low and not considered to be appropriate for design considerations based on typical residential raw wastewater characteristics as well as existing effluent BOD levels at the WWTP. Previous design criteria for the WWTP utilized influent BOD and TSS concentrations of 200 mg/L. Due to water conservation efforts and Building Code requirements for water efficient plumbing on new construction, it is anticipated that per capita flows will go down, causing BOD and TSS concentrations to increase. The design criteria for wastewater influent for the current proposed WWTP improvements therefore used BOD and TSS concentrations of 250 mg/L each.

Table 2-4 Influent Flow Characteristics

		Average	5/25/2016			5/26/2016		
			7:00 AM	12:00 PM	2:30 PM	6:00 AM	11:00 AM	2:00 PM
Alkalinity as CaCO ₃	mg/L	427	410	410	440	430	440	430
Bicarbonate as CaCO ₃	mg/L	427	410	410	440	430	440	430
Carbonate as CaCO ₃	mg/L	ND	ND	ND	ND	ND	ND	ND
Hydroxide as CaCO ₃	mg/L	ND	ND	ND	ND	ND	ND	ND
BOD	mg/L	106	90	71	120	100	170	82
pH		7.9	8.0	8.0	8.0	7.9	7.7	8.0
Settleable Solids	mL/L	2.5	1.9	1.8	3.6	0.9	5.3	1.2
Total Kjeldahl Nitrogen	mg/L	45	42	43	51	42	49	42
TSS	mg/L	130	130	140	110	120	160	120

Influent Electrical Conductivity (EC) and total dissolved solids (TDS) were not sampled, but are assumed to be similar to the effluent levels since EC and TDS are not removed through the biological treatment process.

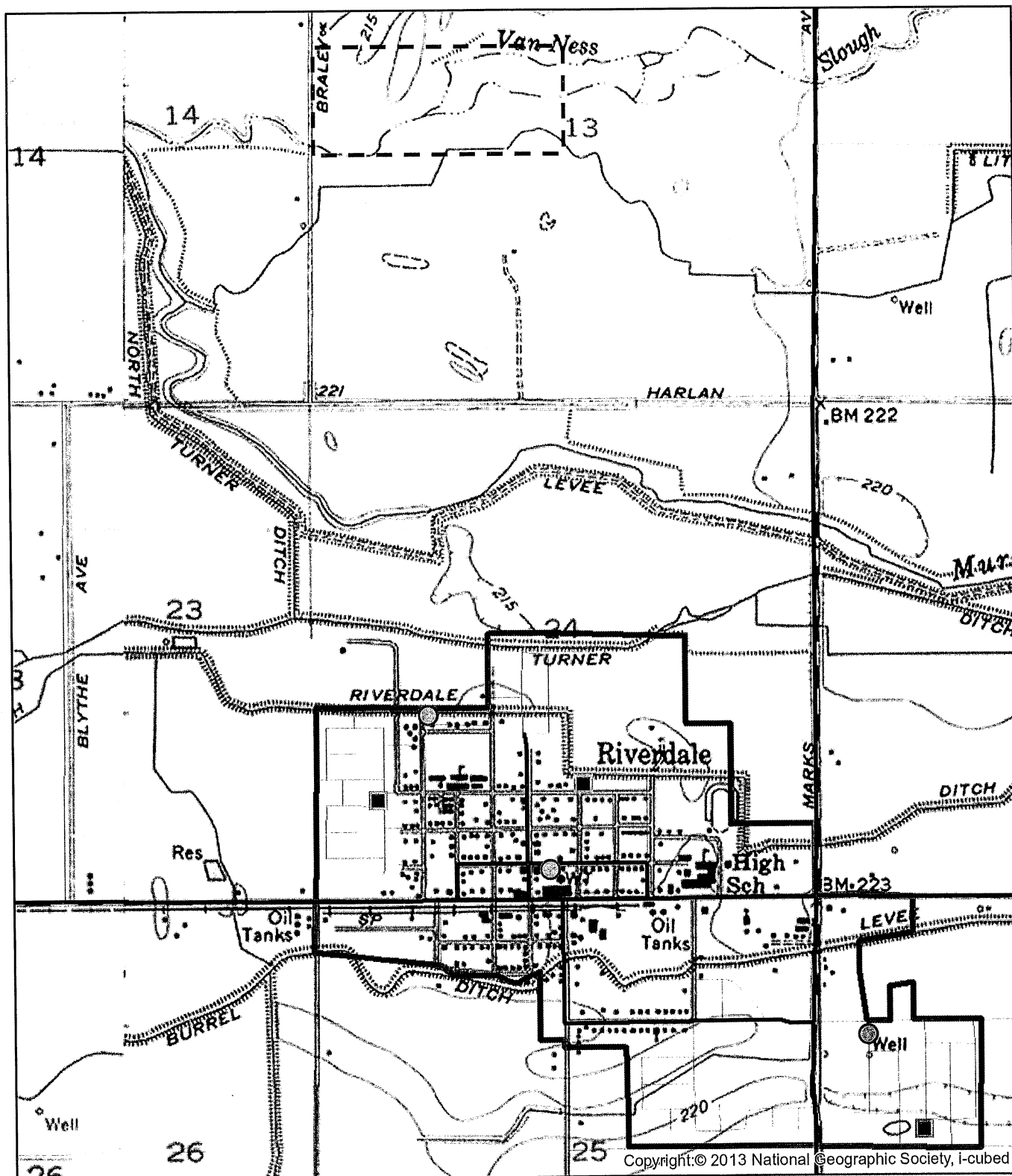
3 Wastewater Treatment Facilities

The topography in Riverdale is flat, with typical ground slopes of less than 0.2 percent. The estimated ground elevation is 215 feet at the WWTP site, based on the Riverdale USGS Quadrangle Map (Figure 3-1). Groundwater elevations were approximately 30 feet (elevation) in the fall of 2015, based on information obtained from the Department of Water Resources (DWR), as shown in Figure 3-2 groundwater contour map. Groundwater elevations have gradually declined since the spring of 2000, when the groundwater elevation was reported to be about 110 feet. DWR groundwater contour maps for the spring of 2016 have shown even further decline to a groundwater elevation of 0-10 feet.

3.1 Existing Treatment System

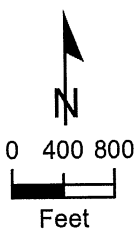
The existing Riverdale PUD WWTP is an aerated lagoon system designed to remove biochemical oxygen demand (BOD) and suspended solids from wastewater. The raw influent is received at the headworks structure, where it passes through a comminutor before going into the complete-mix (CM) aerated lagoon. Effluent from the CM lagoon is distributed to the Oxidation Lagoons, which may be operated in series or in parallel. The Oxidation Lagoons provide for percolation of the effluent into the soil and evaporation to the atmosphere. Effluent from the lagoons may also be applied to the pasture lands adjacent to the WWTP. The existing plant layout is shown in Figure 3-3.

The WWTP has been operating at capacity for several years, and is need of expansion. The existing WWTP has also been unable to achieve the necessary BOD removal. The proposed upgrade of the existing WWTP includes both a capacity expansion and treatment process upgrade to provide more reliable biological treatment.



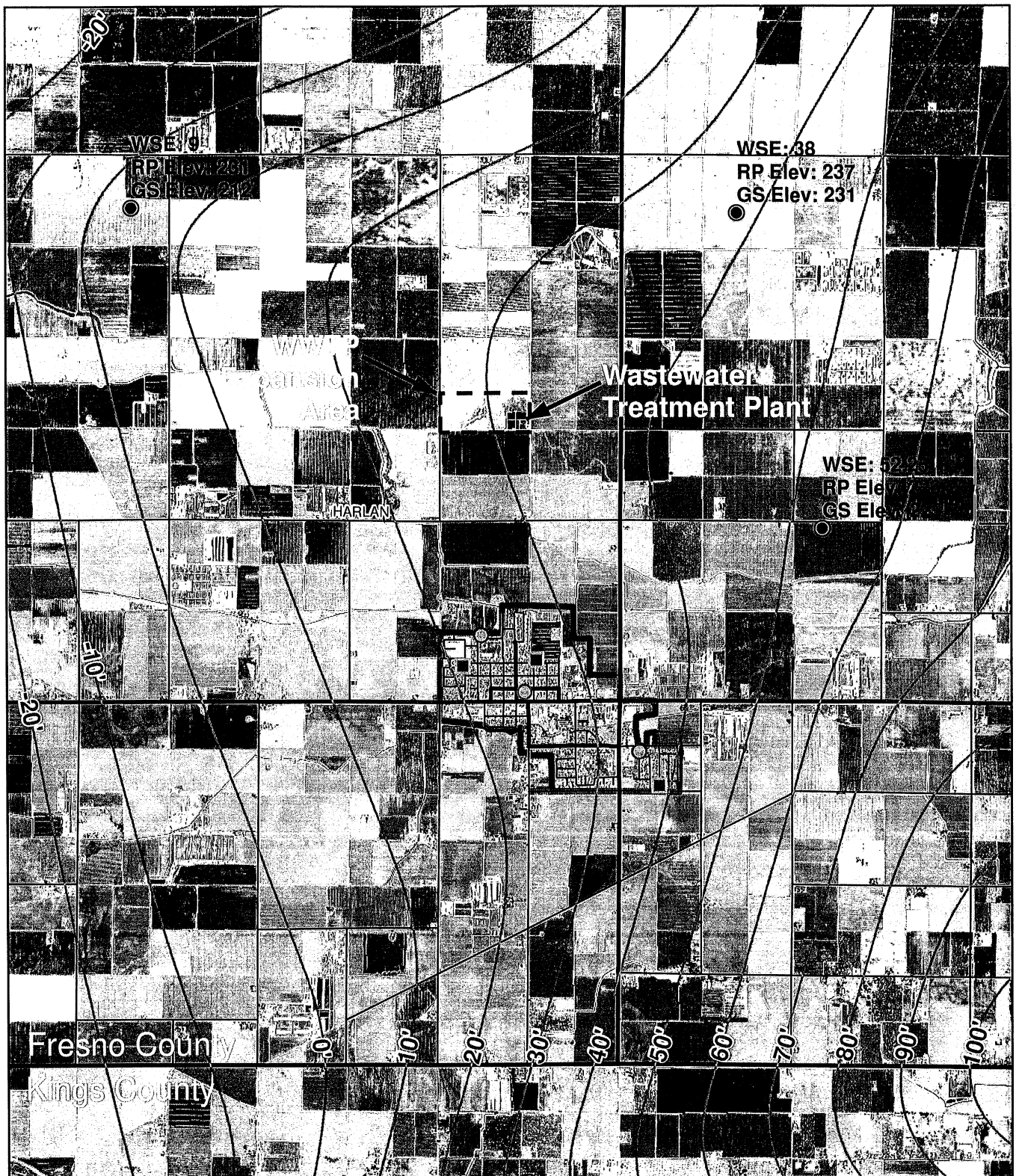
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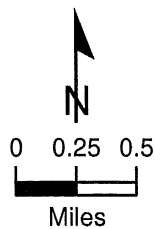


- Wastewater Treatment Plant
- Well
- Lift pump
- Minor Road
- Major Road
- PUD Boundary

Figure 3-1
 Riverdale PUD
 USGS Topo
 Quadrangle Map



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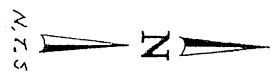
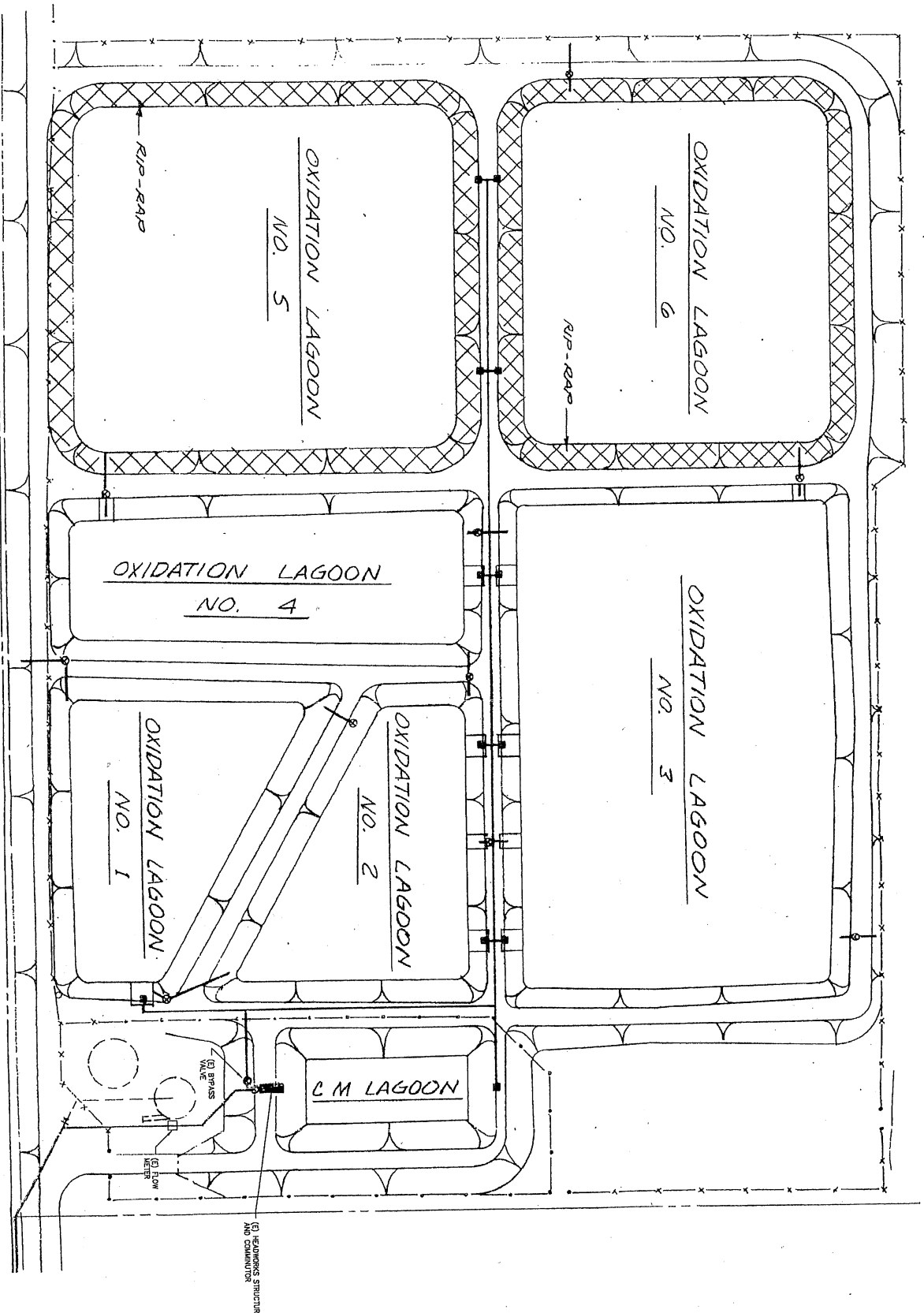


- Wells
- Lift pumps
- Fall 2015 Well Point (DWR GIC)
- Fall 2015 WSE Contour (DWR GIC)
- Minor Road
- Major Road
- ▭ PUD Boundary

GS = Ground Surface
 RP = Reference Point
 WSE = Water Surface Elevation
 DWR = Dept. of Water Resources
 GIC = Groundwater Information Center

Figure 3-2
 Groundwater
 Contour Map

OXIDATION LAGOONS



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3.2 Proposed Treatment System

The current WDRs were issued in 1985, and new WDRs to be issued with the proposed project are expected to have modified effluent limits and greater testing requirements. The new WDRs should be based on a design flow of 0.325 MGD.

Removal of nitrogen is not a requirement of the current WDRs. Based on an antidegradation analysis and correspondence with the RWQCB, removal of nitrogen is not anticipated to be a requirement in the WDRs for the proposed treatment improvements. The Antidegradation Analysis is included in Appendix E.

BOD limits in the current WDRs include a 30-day mean of 40 mg/L and a maximum of 80 mg/L. It is expected that the BOD and TSS limits in the new WDRs will each include a 30-day mean of 30 mg/L and a maximum of 60 mg/L.

The current WDRs state that the maximum EC of the discharge shall not exceed 1,100 μ mhos/cm. However, as discussed with the RWQCB at a meeting on October 1, 2015, and acknowledged in a summary letter dated February 10, 2016, the EC level of the new source wells in the Riverdale community is approximately 1,100 – 1,200 μ mhos/cm. Based on recent correspondence with the RWQCB, RPUD expects that an effluent EC limit will be imposed such that effluent EC shall not exceed the source water EC plus 500 μ mhos/cm.

3.2.1 Influent Pumping

All sewage from the community passes through the lift station located at the corner of Valentine and Stathem Avenues, which pumps it to the WWTP approximately 1.5 miles north of the community through a 10-inch sewer force main. The existing force main to the WWTP was installed in 1958. The pipe material is asbestos-cement and the condition is unknown. According to the District, air valves along the force main have been removed, therefore the existing pipe is susceptible to air accumulating at high points. If only one pump is running (which is the typical operation), the velocities in the force main would be less than 2.0 feet per second, making the pipe susceptible to the buildup of settled solids. The force main is assumed to be past its useful life. The force main is a critical component of the system since it conveys 100 percent of all influent to the WWTP. The existing force main is therefore proposed to be replaced.

A new lift station was recently constructed. This facility includes two new lift pumps, each rated to provide flows of approximately 420 gpm, with space for a third pump. Both pumps operate at constant speed; however, they are equipped with variable frequency drives to allow operators to manually adjust the running speed to optimize pump cycle times. Typically, only one pump will operate at any given time, with a second pump for redundancy. However, during peak flows, both pumps are required, so it is recommended that the third pump be installed to provide redundancy during the peak flow events.

The new sewer force main will be 8-inch or 10-inch diameter, extending from the main sewer lift station to the headworks at the WWTP. The force main will likely be constructed of C900 PVC pipe or fused HDPE pipe.

3.2.2 Headworks

A new headworks structure will be constructed with the new process to accommodate the hydraulic requirements. A new automatically cleaned screen and a bypass channel with a manual bar screen will be installed.

The existing influent flow meter should be replaced with the sewer force main replacement.

3.2.3 Treatment System

The proposed treatment system consists of an aeration pond system that includes three aeration ponds (Aeration Pond No. 1, 2 and 3) that will discharge to evaporation/percolation ponds. Two treatment ponds will be active, with one standby pond. The ponds will be equipped with surface aerators designed to provide aeration and complete mixing. The ponds will be designed to be intensely aerated and mixed, at over 20 horsepower (hp) per million gallons, which will eliminate the risk of temperature overturn or algae growth.

A quiescent zone can be designed to allow settling prior to discharge from the aeration ponds to the disposal ponds. This can be accomplished by the arrangement of the aerators or by installing curtains. The ponds may be designed with two outlets so the settling area can be switched periodically.

The process can be accomplished by constructing two new treatment ponds, each approximately 180 feet by 260 feet, with a volumetric capacity of 2.4 million gallons. Each pond will have a total depth of about 12 to 14 feet, with a minimum of two feet of freeboard. The interior side slope will be about 3:1. The treatment ponds will be lined with a double HDPE liner. A third treatment pond would be constructed to provide redundancy. Aerators would only be installed in two ponds, and would be moved when the ponds are rotated.

The condition of the existing CM lagoon liner will be inspected during construction. If the liner is in good repair, the existing CM lagoon may be retained as an emergency bypass.

The proposed treatment ponds will be raised to allow gravity flow to the disposal ponds. Each pond will be equipped with an estimated five (5) 10 hp aerators.

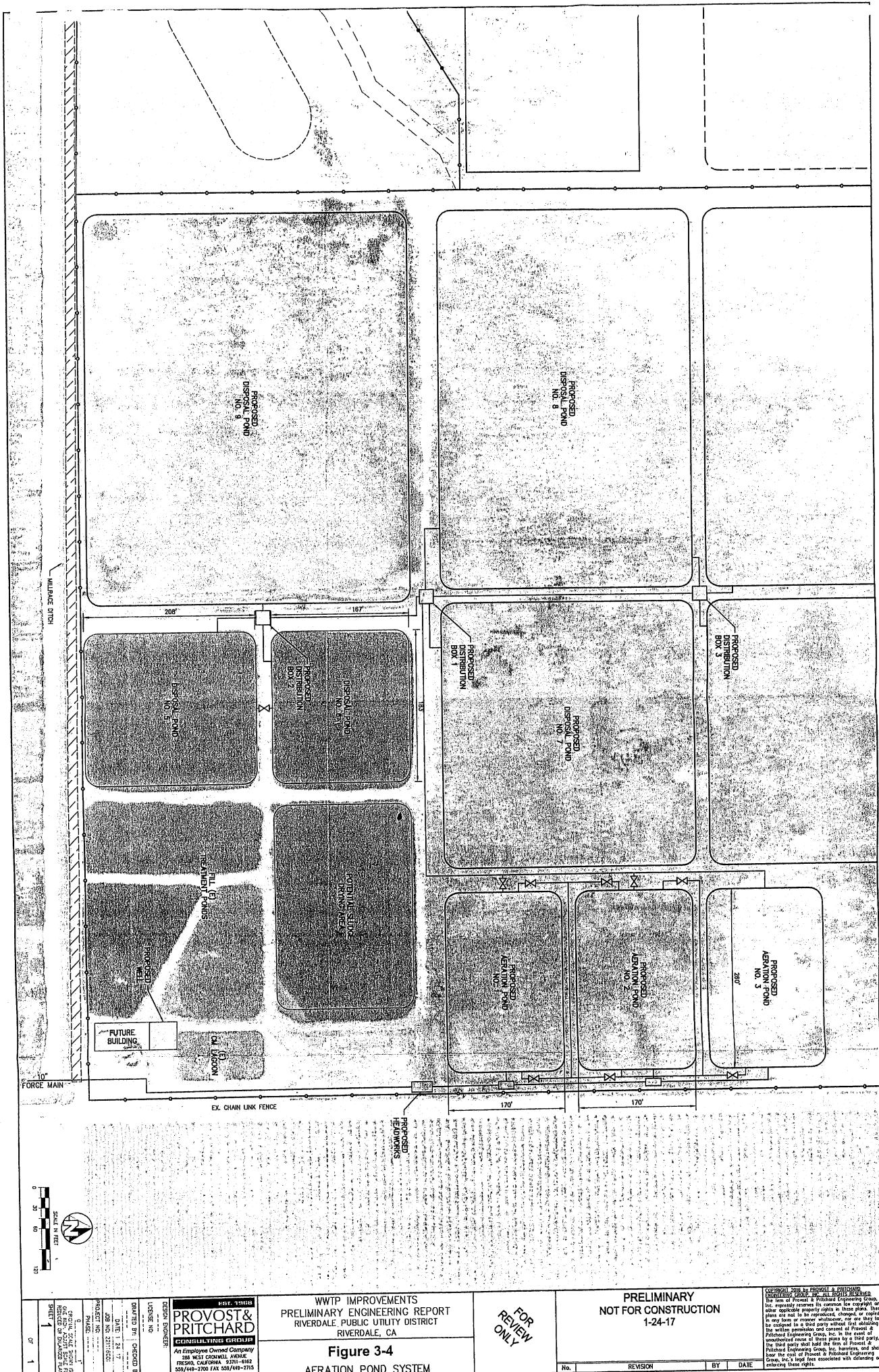
The new treatment facilities can be constructed while the existing WWTP remains entirely in service.

Controls for the facilities are simple and based on a timer. The monitoring requirements will be similar to current operation. The addition of more surface aerators requires more mechanical maintenance.

A preliminary layout and process flow diagram are included in Figure 3-4 and Figure 3-5.

The ponds will be designed with a minimum of 2 feet freeboard, and the discharge will flow by gravity to the disposal ponds. Sufficient drop between the treatment ponds and the disposal ponds will be allowed so that overfilling or overflow will not be an issue. A preliminary hydraulic profile is shown in Figure 3-6. Pond design and hydraulic profile will be updated during design.

Operation and maintenance procedures will include monitoring pond levels and effluent quality in accordance with the WDRs as promulgated by the RWQCB; checking aerator operation and maintaining or replacing aerators as necessary; and periodically checking sludge levels and dredging as necessary. Depth to shallow groundwater below the site is approximately 220 feet (refer to Appendix E - Antidegradation Analysis).



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CONSULTING GROUP, INC.
An Employee Owned Company
280 WEST CROCKWELL AVENUE
FREMONT, CALIFORNIA 94531-6162
508/449-2700 FAX 508/449-2715
www.ppv.com

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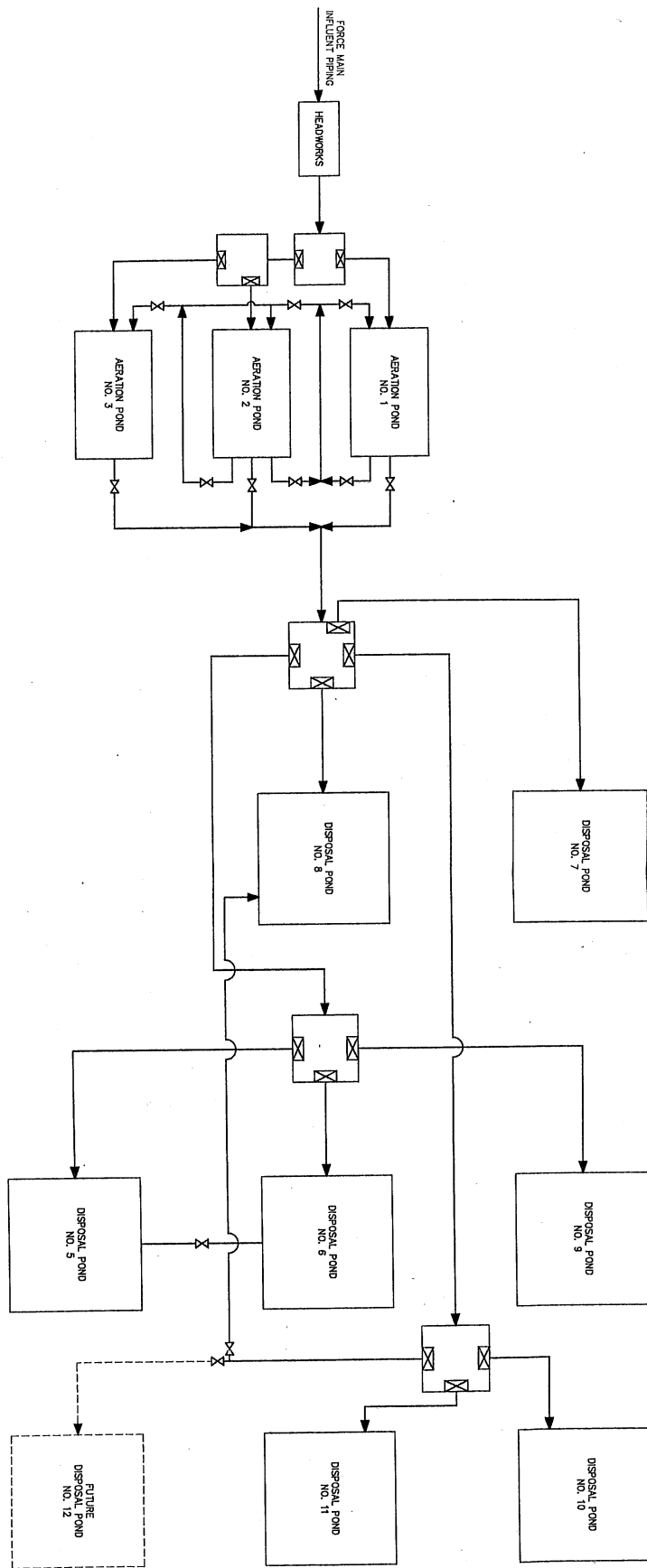
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Figure 3-4
AERATION POND SYSTEM

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Figure 3-5

PRELIMINARY PROCESS FLOW DIAGRAM

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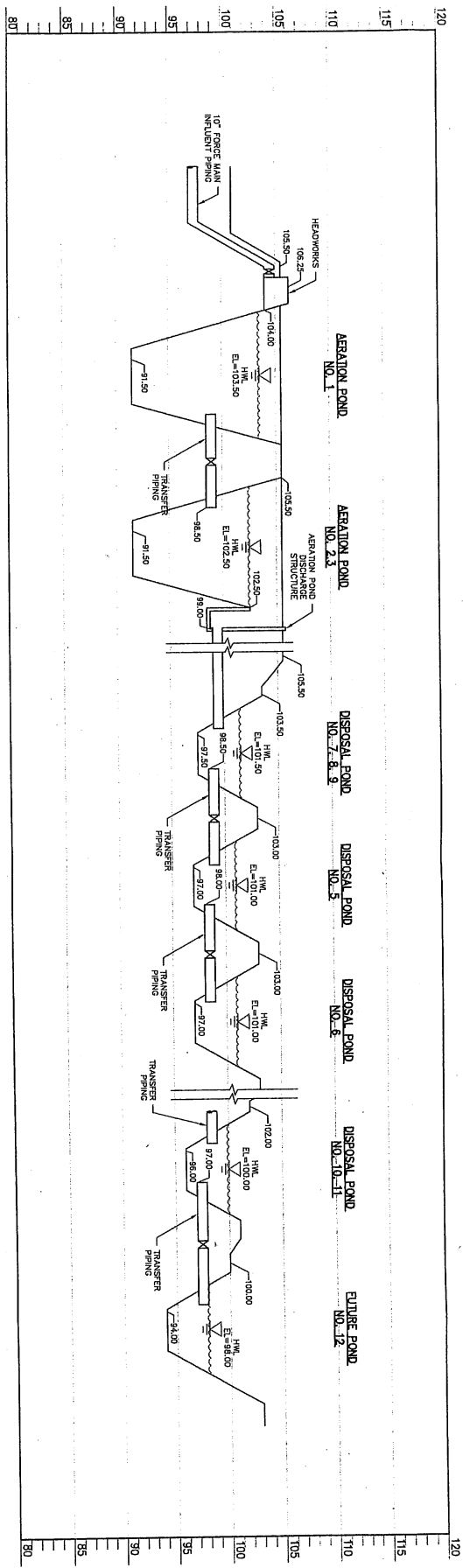
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286 WEST CROMWELL AVENUE
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DESIGNER _____
CHECKED BY _____
DATE 1-24-17
PROJECT NO. 22111600
SHEET 1 OF 1
ON-SITE SCALE 1"=20' HORIZONTAL
ON-SITE SCALE 1"=10' VERTICAL
RECORDS & PLANNING



WARNING
POWER LINES
OVERHEAD



WARNING
POWER LINES
OVERHEAD



1/24/2017 12:21:04 PM C:\Users\pud - 2211\Documents\Westwater Treatment Plant\DWG\XREF\1\ PRELIMINARY HYDRAULIC PROFILE.dwg - Stephen Gilvez

Figure 3-6
PRELIMINARY HYDRAULIC PROFILE

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3.2.4 Effluent Disposal Facilities

3.2.4.1 Storage and Disposal Ponds

Existing disposal facilities include evaporation and percolation that occurs in the existing oxidation lagoons. Effluent from the oxidation lagoons is released to the adjacent reclamation area for disposal, as needed. The reclamation area is not currently harvested.

Effluent from the new treatment facilities will be discharged into a series of new storage and disposal ponds. Construction of the storage and disposal ponds will be phased, since 0.325 MGD of disposal capacity will not be necessary for several years. The first phase would include storage and disposal capacity for up to 0.275 MGD, which should be sufficient for more than ten years. Some of the existing treatment ponds will be converted to storage and disposal ponds, as shown in Table 3-1, and the existing reclamation area will be replaced, since it is not currently being harvested.

Approximately 21.7 acres of new storage ponds will initially be constructed in order to meet an increased capacity to 0.275 MGD. As flows increase, future pond(s) will be constructed for discharge of the ultimate WWTP design capacity of 0.325 MGD flow. While disposal area for the entire treatment capacity is not anticipated to be necessary for at least 10 years, provisions will be constructed for these future facilities in the current project. The proposed storage ponds will be located to the west and north of the existing treatment lagoons. In addition, the existing Oxidation Lagoons 5 and 6 will be converted to storage and disposal ponds. Effluent from the proposed treatment facilities will flow by gravity into the proposed storage and disposal ponds.

Total storage volume of approximately 96 ac-ft and 24 acres (surface area) of disposal ponds will be provided for effluent disposal for 100-year return condition at effluent flows of 0.275 MGD as calculated in water balance spreadsheet (Appendix C). A summary of proposed storage pond sizes is shown in Table 3-1.

For the ultimate WWTP capacity of 0.325 MGD, an additional 4.6 acres of pond surface area will be required to supplement the evaporation and percolation from the initial phase ponds (see Appendix C). See Figure 3-7.

Table 3-1 Proposed Storage and Disposal Ponds

Pond	Volume MG	Surface Area Acre	Water Depth ft
(E) Oxidation Lagoon 5 (Proposed Disposal Pond 5)	1.7	1.32	4
(E) Oxidation Lagoon 6 (Proposed Disposal Pond 6)	1.4	1.09	4
Proposed Pond 7	4.2	3.2	4
Proposed Pond 8	6.0	4.6	4
Proposed Pond 9	8.0	6.1	4
Proposed Pond 10	4.2	3.2	4
Proposed Pond 11	6.0	4.6	4
Total (Proposed)	31.5	24.1	
Future Pond 12	6.0	4.6	4
Total (Future)	37.5	28.7	

Prior to conversion of existing oxidation ponds to disposal ponds it is recommended that sludge be removed from each pond and ponds be disked to improve percolation.

A large volume of earthwork will be needed to construct the new storage and disposal ponds. It may be possible to raise the existing berm structure on the ponds to balance the cut and fill. However, the demand of soil for development may make soil export an economical option. Soil could be stored on the additional 33 acres owned by the District to the west of the WWTP.

3.2.4.2 Water Balance

The effluent storage and disposal pond sizing is based on water balance calculations, which include influent flows, precipitation, percolation, and evaporation. The ponds will be designed with a minimum of 2 feet freeboard under all conditions. Projected monthly water balances were developed to demonstrate that adequate containment capacity for both the average rainfall year and the 100-year return period total annual precipitation.

The design influent flow is the maximum design flow of the proposed WWTP, which is 0.325 MGD. However, since it will be several years before flows reach this volume and interim design flow for the disposal ponds of 0.275 MGD has been considered.

Evaporation data used in the water balance calculations is based on evapotranspiration data from the Cal Poly Irrigation Training & Research Center (ITRC). The evapotranspiration data from the ITRC was modified to be used for large ponds. It is assumed that large ponds, such as those proposed, will have evapotranspiration of about 125% of the pan evaporation values.

Precipitation data used in the water balance calculations is based on the Western Regional Climate Center, Southern California Climate Summary for the Fresno area. These include an average year rainfall as well as a 100-year return period rainfall.

A percolation rate of 0.33 inches per day has been assumed for the water balance calculations. The existing unlined oxidation ponds have not had sludge removed nor have they been disked to maintain percolation. The percolation rate from these ponds is not monitored, nor would it be considered to be representative of the potential percolation from a maintained pond. A percolation rate of 0.33 inches per day is considered to be a conservative estimate. Percolation tests in the proposed disposal pond area should be conducted during the design phase, which should give a better estimate of percolation rates.

Water balance calculations are included in Appendix C.

3.2.4.3 Evaluation of Reclamation

Use of recycled water for irrigation could alternatively be considered for disposal in the future. The secondary treatment level and the high EC levels in the effluent will limit the type of crops that could be used. Alfalfa irrigation could be suitable for use of secondary effluent for irrigation with the current effluent EC levels, although any increase in EC levels could impact the crop yield. Salt tolerance for alfalfa is approximately 2,000 $\mu\text{mhos}/\text{cm}$. If recycled water use is pursued, an agreement would need to be developed with a nearby farmer who grows alfalfa, hay, or other fodder crops for animals not producing milk for human consumption. If necessary, an effluent pump could be used to pump effluent from the ponds to the reclamation area. Use of recycled water for irrigation is not anticipated to be practicable at this time due to the relatively low effluent flows, and since there is no current agreement or demand for the use of recycled water.

A letter dated November 29, 2016 was sent to property owners within a one-mile radius of the WWTP, who currently grow alfalfa, hay, or other fodder crops (see Appendix D). The letter requested response from property owners who are interested in the potential of using treated effluent from the RPUD WWTP for reclamation. As of the date of this report, response from one property owner was received. The property owner expressed interest in leasing the RPUD owned property adjacent to the WWTP to grow forage crops utilizing recycled water. The property owner was not interested in using the effluent on his existing crops, due to the high EC levels in the effluent. The RPUD is not interested in creating a new demand on the RPUD property at this time. The RPUD Board of Directors suggested following up with a second property owner who had expressed interest in using reclaimed water in the past. The second property owner had the same concerns with the high levels of EC in the effluent, and was therefore not interested in using recycled water at this time.

Use of effluent from the RPUD WWTP for reclamation is therefore not a viable option for this project.

3.2.5 Solids Handling

Sludge may be dredged and removed from the treatment ponds as necessary. The removed sludge can be mechanically dried or dried in a designated sludge drying area and hauled offsite.

It is unknown if or when sludge has been removed from the existing treatment ponds. It is recommended that sludge levels in the proposed treatment ponds be checked every two to three years. In addition, if a decline in effluent quality is noticed, sludge levels should be checked. When necessary, due to the sludge level and/or reduced treatment performance, sludge will be removed. With two treatment ponds, one can be temporarily taken out of service for dredging.

3.2.6 Electrical Facilities

The existing electrical facilities are old and likely insufficient for the proposed WWTP improvements. The electrical facilities will be replaced to accommodate the demands of the proposed project.

The existing WWTP does not have a Supervisory Control and Data Acquisition (SCADA) system. It is recommended that SCADA be considered with the WWTP upgrades to provide remote monitoring and alarm capabilities, as well as providing automatic reporting of critical information.

3.2.7 Supply Water

The WWTP does not currently have supply water for wash down of the facilities. Providing service from the District's potable water system would be cost prohibitive since it would require installation of about 1.5 miles of pipeline from the community to the WWTP.

A new well is proposed to be installed at the WWTP to provide water for rinsing and wash down activities. Hose bibs would be provided near the headworks and treatment ponds for wash down of those facilities. Since the proposed well will not be intended for drinking water purposes, appropriate signage will need to be posted.

4 Characterization of Water Quality

4.1 Source Water Quality

The RPUD Well 6, which was constructed in 2015, is the only active well. A new Well 7 is planned for construction in 2017, and is anticipated to have similar water quality. RPUD Wells 4 and 5 are standby wells for emergency use, but have been taken out of active service due to arsenic contamination.

Source water quality from RPUD Well 6 is summarized in Table 4-1.

Table 4-1 Source Water Quality

Constituent	Units	Result
Total Dissolved Solids	mg/L	610
Electrical Conductivity	µmhos/cm	1100
Total Kjeldahl Nitrogen as N	mg/L	NS
Ammonia Nitrogen as N	mg/L	NS
Nitrate as NO ₃	mg/L	ND
pH	pH Units	8.6
Alkalinity as CaCO ₃	mg/L	290
Hardness (Total) as CaCO ₃	mg/L	8.4
Bicarbonate	mg/L	310
Carbonate	mg/L	21
Calcium	mg/L	2.8
Magnesium	mg/L	0.34
Chloride	mg/L	180
Potassium	mg/L	ND
Sodium	mg/L	250
Sulfate	mg/L	ND
Arsenic	µg/L	6
Iron	µg/L	ND
Manganese	µg/L	ND

ND = Not Detected

NS = Not Sampled

4.2 Influent Wastewater Quality

Influent flows to the WWTP are primarily from residential uses, with some commercial and institutional connections. Influent flow characteristics are summarized in Table 2-4. The sampling results for influent BOD and TSS are very low and not considered to be appropriate for design considerations based on typical residential raw wastewater characteristics as well as existing effluent BOD levels at the WWTP.

4.3 Treated Effluent Quality

The wastewater effluent produced by the proposed RPUD WWTP is classified as undisinfected secondary effluent, and will be designed to achieve effluent characteristics no greater than 30 mg/L BOD and no greater than 30 mg/L TSS. Due to high salinity levels in the RPUD supply water, the WWTP effluent has EC levels of about 1,540 µmhos/cm, on average. Projected effluent quality is summarized in Table 4-2.

Table 4-2 Projected Effluent Quality

Constituent	Units	Result
Biochemical Oxygen Demand ^a	mg/L	30
Chemical Oxygen Demand	mg/L	NS
Settleable Matter	mL/L	NS
Total Suspended Solids ^a	mg/L	30
Total Dissolved Solids ^b	mg/L	900
Electrical Conductivity ^c	µmhos/cm	1540
Total Kjeldahl Nitrogen as N ^d	mg/L	45
Ammonia Nitrogen as N	mg/L	NS
Nitrate Nitrogen as N ^e	mg/L	<1
pH	pH Units	6.5-8.5
Alkalinity as CaCO ₃	mg/L	NS
Hardness (Total) as CaCO ₃	mg/L	NS
Bicarbonate	mg/L	NS
Carbonate	mg/L	NS
Calcium ^b	mg/L	3
Magnesium ^b	mg/L	1
Chloride ^b	mg/L	250
Potassium ^b	mg/L	<1
Sodium ^b	mg/L	250
Sulfate ^b	mg/L	<1
Aluminum ^b	µg/L	<1
Arsenic ^b	µg/L	6
Copper ^b	µg/L	<1
Iron ^b	µg/L	<1
Lead ^b	µg/L	<1
Mercury ^b	µg/L	<1
Manganese ^b	µg/L	<1
Zinc ^b	µg/L	<1

a. Projected effluent concentrations are estimated based on proposed treatment design criteria.

b. Projected effluent concentrations are estimated based on source water quality.

c. Projected effluent concentrations are based on existing effluent samples since Well 6 has come online.

d. Projected effluent concentrations are estimated based on existing influent quality.

e. Projected effluent concentrations are estimated based on existing effluent quality.

5 Inflow and Infiltration

The sewer collection system consists primarily of 6-inch to 10-inch diameter asbestos-cement gravity sewer, which collects at the lift station located at the corner of Valentine and Stathem Avenues. The sewer collection system was originally constructed in 1958.

All sewage from the community passes through the lift station located at the corner of Valentine and Stathem Avenues, which pumps it to the WWTP, approximately 1.5 miles north of the community, through a 10-inch sewer force main. The existing force main to the WWTP was installed in 1958. The pipe material is asbestos-cement.

A new lift station was recently constructed. This facility includes two new lift pumps, each rated to provide flows of approximately 420 gpm, with room for a third pump. Typically, only one pump operates at any given time, with a second pump for redundancy. Installation of the third pump is proposed to provide redundancy during peak flows.

Based on influent flow data presented in Table 2-3, inflow and infiltration does not appear to be an issue for the RPUD. The average monthly flows from January 2011 to December 2016 are summarized in Table 5-1. This shows that the maximum month influent flows received at the WWTP (0.238 MGD) are less than 4 percent increase over the annual average flows (0.23 MGD). This is within acceptable fluctuations in normal wastewater flow, without significant impacts from inflow or infiltration.

Table 5-1 Average Monthly Influent Flows 2011 to 2016

Month	Average Flow by Month (MGD)
January	0.227
February	0.228
March	0.225
April	0.223
May	0.222
June	0.222
July	0.223
August	0.228
September	0.228
October	0.232
November	0.230
December	0.221
Annual Average	0.226
Maximum Month*	0.232

* The maximum month is averaged over the data period, since average and maximum flows have gradually decreased from 2011 to 2016.

6 Preventative and Contingency Measures

6.1 Reliability Provisions

6.1.1 Lift Station

The lift station pumps are each capable of handling current average demands with a single pump. A second pump is included for redundancy should pump failure occur. The lift station is configured to accommodate a third pump. Installation of the third pump is proposed to provide redundancy during peak demands.

The lift station pumps alternate as lead and lag at each start cycle. The pumps are called on and off by level sensors in the wet well. The following alarms are reported to the SCADA system:

- Lift Station Wet Pit high-high level alarm
- Pressure transducer high-high pressure alarm
- Lift Station power fail
- Uninterruptable Power Supply (UPS) fail
- Remote Terminal Unit (RTU) power fail
- VFD fail

Provisions are available to accommodate a portable emergency standby generator in the event of power failure.

6.1.2 WWTP

Reliability provisions will be provided at the WWTP in case of potential equipment failure.

The headworks will be equipped with a bypass channel and manual bar screen that can be used if the automatically cleaned screen is temporarily out of service. The operator would manually operate gates at the headworks structure to redirect flow through the bypass channel when this is necessary. This will be an infrequent operation.

The proposed WWTP improvements will include three (3) treatment ponds. Two treatment ponds will be active, with one standby. The treatment ponds are designed to normally operate in series, although parallel operation can be accommodated. Typically only two ponds will be in operation, with the third pond available for redundancy, in case one of the ponds needs to be taken out of service. The treatment ponds are designed to achieve BOD removal to 30 mg/L in the winter, when biological removal is reduced due to the lower temperatures.

For additional reliability, the existing CM lagoon may be retained as an emergency bypass. The condition of the CM lagoon liner will need to be inspected during construction to determine if this is viable.

There will be multiple aerators in each treatment pond, so treatment can be sustained when one needs to be taken out of service temporarily for maintenance or replacement.

There are multiple disposal ponds, which are sized for a total disposal capacity to accommodate a 100 year rainfall event, so ponds can be periodically emptied and disked during normal or dry years to maintain percolation capacity.

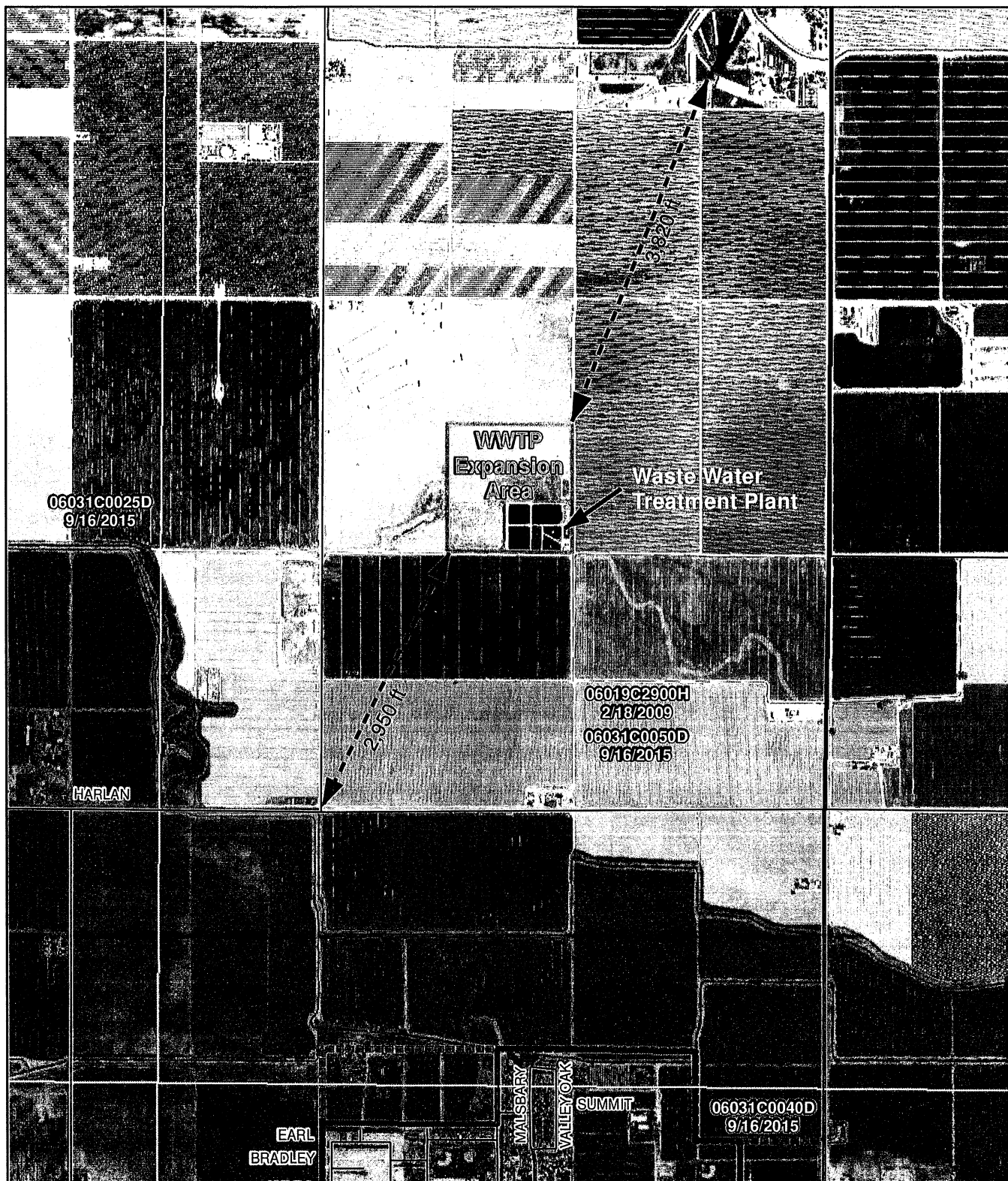
6.2 Spill and Discharge Control

The proposed WWTP is a pond based system. All treatment and disposal ponds will be designed with a minimum 2 feet of freeboard. In addition, disposal facilities will be designed for a 100 year rainfall event. With these provisions, no spills or accidental discharges are anticipated.

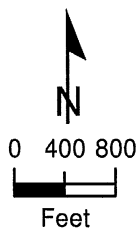
6.3 Flood and Frost Protection

The project area is outside of the FEMA 100 year flood zone, as shown on Figure 6-1. The WWTP is within the area designated as FEMA Flood Zone X, area of minimal flood hazard.

Seasonal low temperatures typically do not go below about 35 °F in the winter. While some frost may occur, it is not anticipated to cause any structural issues. With the complete mixing in the treatment ponds, frost will not cause operational issues.



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- DFIRM Panel Number/Date
- 100 Year Flood Zone
- PUD Boundary
- SOI Boundary

Figure 6-1
 100 Year Flood Zone
 FEMA DFIRM Map

7 Operation and Maintenance Procedures

7.1 O&M Procedures

The treatment ponds can be operated in series or in parallel. It is recommended that normal operation is in series. Two treatment ponds should be in use at any given time. A third treatment pond is provided for redundancy. For periodic maintenance purposes, and to remove sludge, one treatment pond can be taken out of service, and the third pond can be placed in service to maintain treatment performance.

The efficiency of the treatment system is affected by temperature, and it is therefore important to keep two ponds in operation during winter months. During summer months, the treatment efficiency should increase. Pond maintenance activities that require pond downtime should be conducted in the summer.

Operation and maintenance procedures will include monitoring pond levels and effluent quality in accordance with the WDRs as promulgated by the RWQCB; checking aerator operation and maintaining or replacing aerators as necessary; and periodically checking sludge levels and dredging as necessary.

Periodic maintenance of the disposal ponds is necessary to remove sediment build-up and maintain adequate percolation rates. Percolation ponds shall be maintained and periodically disked to sustain the percolation capacity. During normal or dry rainfall years, a single disposal pond can be taken out of service for maintenance and disking. Ponds should be rotated such that each year a different disposal pond can be disked.

Headworks facilities should be observed and maintained as necessary, and the removed grit and screenings hauled offsite.

The following is a preliminary maintenance and inspection schedule that the operator could use to establish an operation and preventative maintenance program to maintain peak performance.

Daily:

- Check aerator operation. Inspect mooring cables
- Check automatic screen operation. Inspect influent channel for debris that might interfere with normal operation
- Test dissolved oxygen in the aeration ponds and pond effluent
- Check aerator run times and adjust as necessary
- Check average daily flows from influent flow meter
- Complete daily operation record

Weekly:

- Wash down treatment pond sides and remove floating debris
- Test plant effluent BOD, TSS, pH, and EC levels (frequency and constituents tested will be in accordance with the WDRs)
- Complete weekly operation record

Monthly:

- Test raw influent BOD, TSS, and settleable solids (frequency and constituents tested will be in accordance with the WDRs)

- Check condition of treatment and disposal ponds, embankments, and overall WWTP site
- Complete monthly operation record

Annually:

- Check aerator performance. If there is a noticeable decrease in mixing intensity or flow output from an aerator, pull the aerator, inspect, and repair or replace as necessary
- Exercise all transfer valves and diversion valves/gates
- Complete annual maintenance record.

7.2 Operator Certification

The RPUD currently has two operators who operate the water and sewer systems. David McIntyre (#10338) is a Grade II operator, and Vincent Romero (#41849) is a Grade I operator. A Grade II operator certification will be required to operate the proposed facility.

On average, the operators spend approximately 15-20 hours per week at the WWTP, depending on the time of year and maintenance needs.

8 Local Conditions

8.1 General Conditions

The RPUD is located approximately 25 miles south of the City of Fresno, in rural Fresno County, California. A vicinity map is included as Figure 1-1. Riverdale is a predominantly agricultural community with no large industrial businesses present. The community had a population of 3,153 in the year 2010, according to US Census data. The current population is estimated to be 3,197 based on the current number of residential sewer service connections (864) and approximately 3.7 persons per household. Figure 1-2 shows a map of the service area.

Riverdale is located approximately four miles up-gradient from the north fork of the Kings River. No rivers or streams flow through the community. The topography in Riverdale is flat, with typical ground slopes of less than 0.2 percent. The estimated ground elevation is 215 feet at the WWTP site, based on the Riverdale USGS Quadrangle Map (Figure 3-1). Groundwater elevations were approximately 30 feet (elevation) in the fall of 2015, based on information obtained from the Department of Water Resources (DWR), as shown in Figure 3-2 groundwater contour map. Groundwater elevations have gradually declined since the spring of 2000, when the groundwater elevation was reported to be about 110 feet. DWR groundwater contour maps for the spring of 2016 have shown even further decline to 0-10 feet elevation.

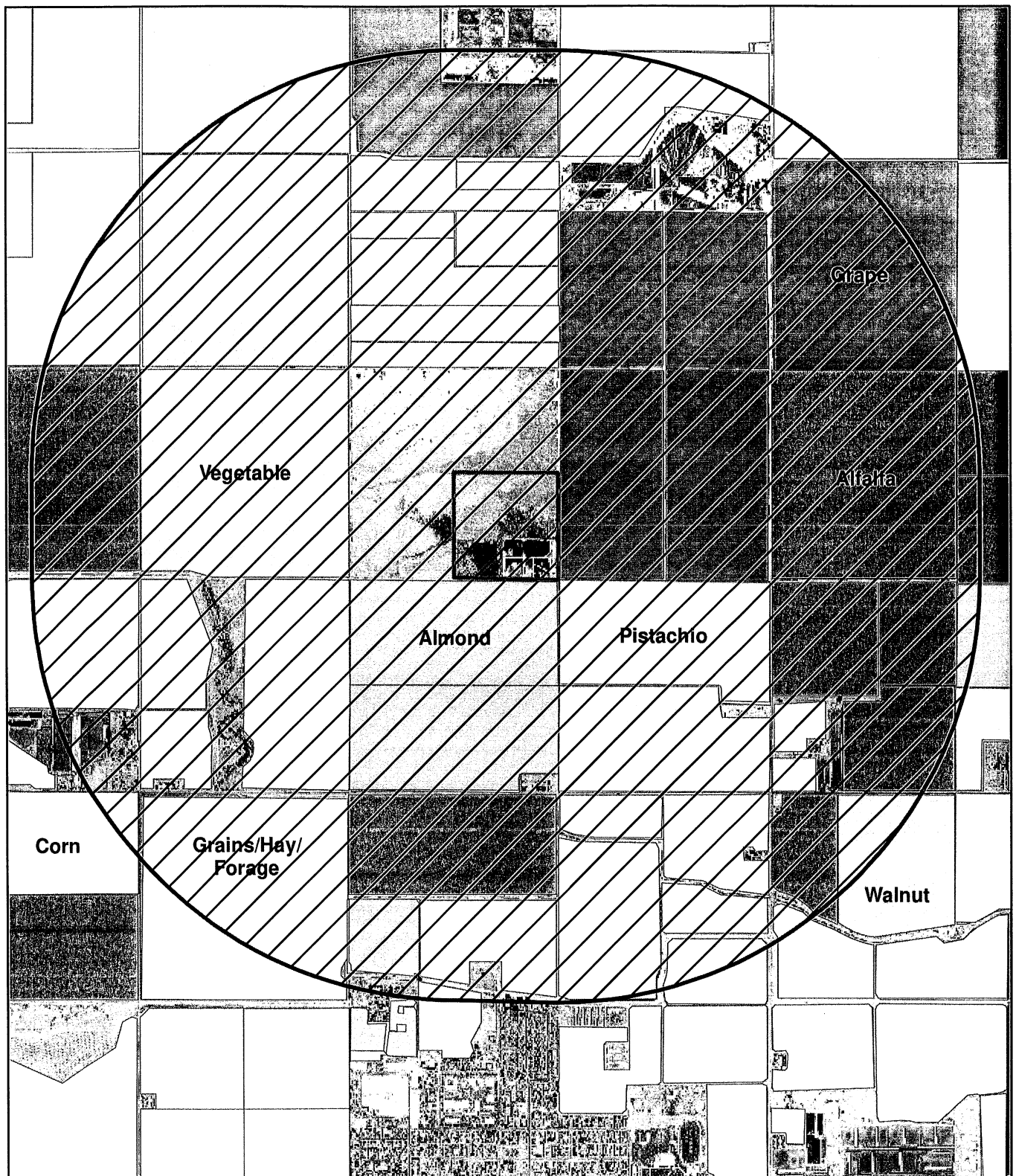
Climate is typical of the San Joaquin Valley. Summers are hot and arid. Little if any rain falls from June through August. Winters are marked by long periods of fog and overcast punctuated by periodic rains. Annual precipitation averages about 10 inches, with nearly all of that falling between November and April. Prevailing winds in the area are from the northwest and average 5-8 mph. Storm winds are typically from the southeast during winter months and are characterized by velocities up to 30 mph.

The WWTP is located east of the Brawley Avenue alignment and south of Cerini Avenue, approximately 1.5 miles north of the Riverdale community. The existing WWTP site is on APN 053-090-011, located in Section 13, Township 17 South, Range 19 East, MDB&M, at Latitude 36° 27' 25" North, Longitude 119° 52' 00" West. An additional 40-acre tract of land was acquired by the District, which adjoins the west boundary of the existing property and consists of the Southwest ¼ of the Northwest ¼ of Section 13. The Van Ness Slough remnant is located in the southeast corner of this additional 40-acre property. The Van Ness Slough remnant will be avoided, limiting the 40-acre site to approximately 33 acres of useable land.

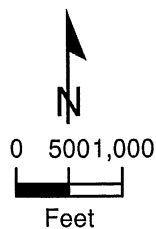
The area around the RPUD WWTP is primarily agricultural. Typical crops grown nearby include alfalfa, almonds, corn, grains/hay/forage, grapes, pistachios, vegetables, and walnuts, as shown in Figure 8-1.

Irrigation water sources in the area include groundwater and surface water supplies. The nearest surface water bodies include the Little Millrace Ditch, which runs east and west along the southern boundary of the WWTF and a remnant of the Van Ness Slough located on the southeast corner of the western 40-acre use area.

The Little Millrace Ditch is an irrigation ditch owned and operated by the Liberty Millrace Irrigation District. The proposed improvement area will be graded such to contain surface drainage onsite and will not be discharged to the canal.



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Crop Category

- Alfalfa
- Almond
- Corn
- Grains/Hay/Forage

- Grape
- Pistachio
- Vegetable
- Walnut

Figure 8-1
 Ag Commissioner
 Crop Map

8.2 Soils

Soils around the RPUD WWTP site are primarily characterized by Chino loam, with some Traver fine sandy loam, as shown on Figure 8-2.

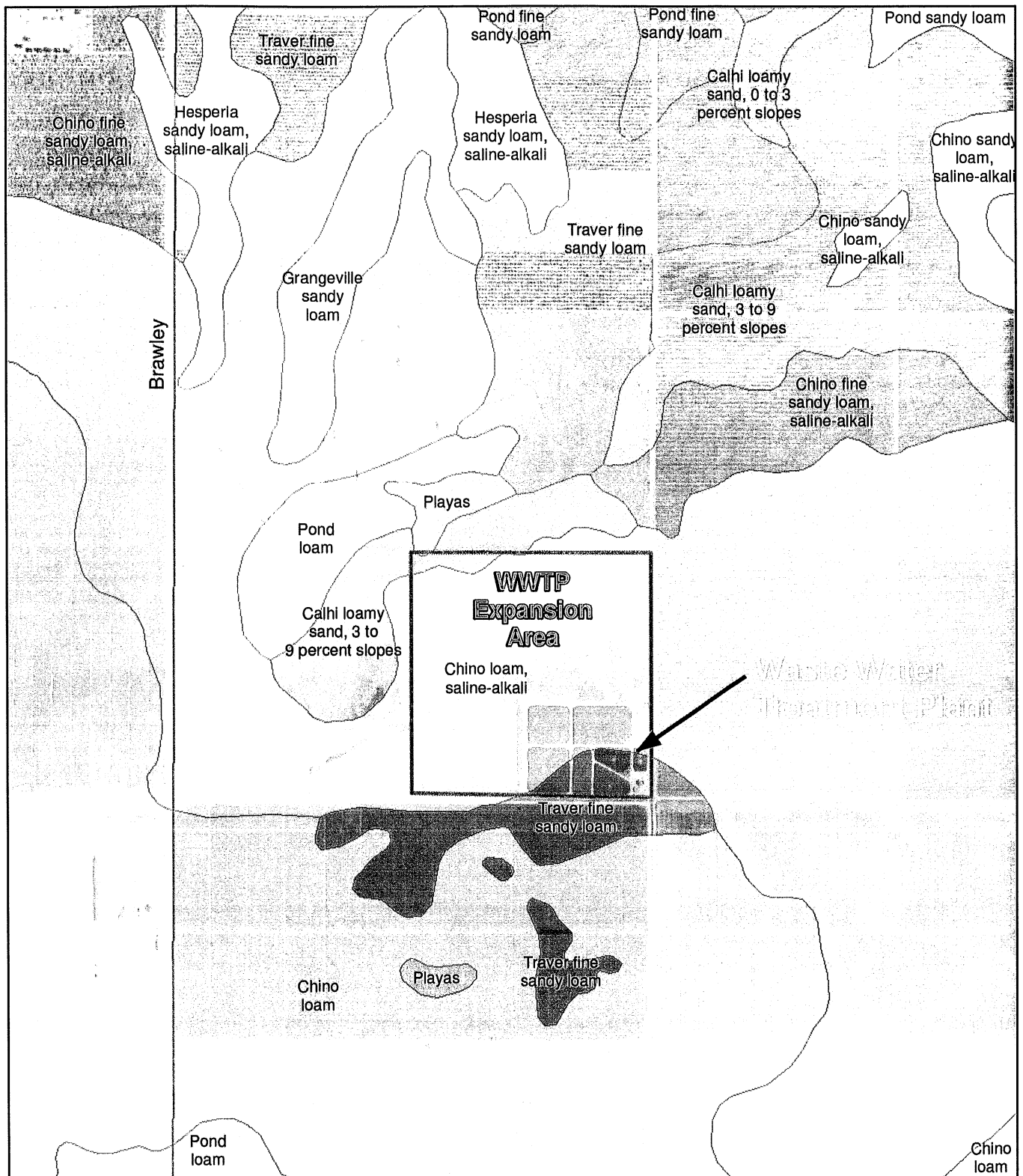
The percolation rate used in the water balance calculations (0.33 inches/day) was taken from a conservative interpretation of earlier data and experience in the vicinity. No project-specific data is available.

8.3 Hydrogeology and Groundwater

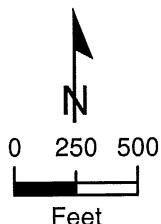
The hydrogeology and groundwater conditions are described in detail in the Antidegradation Analysis included in Appendix E. Flow direction has been variable, but predominantly in the westerly to northwesterly direction. The average groundwater gradient in the area is about 0.0021.

The average groundwater elevation has dropped about 100 feet at the site, from an elevation of 110 feet in spring 2000, to an elevation of 0-10 feet in spring 2016. See the Antidegradation Analysis for more information groundwater conditions.

There are no existing monitoring wells at the site.



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- fine sandy loam
- loam
- loamy sand
- playas
- sandy loam

Figure 8-2
 Soil Texture
 NRCS Soil Survey

9 Antidegradation Analysis

The Antidegradation Analysis is included in Appendix E.

10 Industrial Storm Water Permit

This section is not applicable. The RPUD is exempt from the Industrial Storm Water Permit since the design flow is less than one million gallons per day.

11 General WDRs for Sanitary Sewer Systems

The RPUD sanitary sewer system is covered under the Statewide General WDRs for Sanitary Sewer Systems (Order No. 2006-0003-DWQ).

12 Monitoring Wells

No monitoring wells exist at the site. This section is therefore not applicable.

If monitoring wells are required to be installed at the site, locations and depths would be coordinated with the RWQCB. New monitoring wells would be constructed in accordance with appropriate Department of Water Resources Well Standards.

Appendix

Appendix A
Form 200

**CALIFORNIA ENVIRONMENTAL
PROTECTION AGENCY**

 State of California
Regional Water Quality Control Board

**APPLICATION/REPORT OF WASTE DISCHARGE
GENERAL INFORMATION FORM FOR
WASTE DISCHARGE REQUIREMENTS OR NPDES PERMIT**

I. FACILITY INFORMATION
A. Facility:

Name: Riverdale Public Utility District			
Address: P.O. Box 248			
City: Riverdale	County: Fresno	State: CA	Zip Code: 93656
Contact Person: Vincent Romero		Telephone Number: (559) 867-3838	

B. Facility Owner:

Name: Riverdale Public Utility District			Owner Type (Check One) 1. ☐ Individual 2. ☐ Corporation 3. ☒ Governmental Agency 4. ☐ Partnership 5. ☐ Other: _____	
Address: P.O. Box 248				
City: Riverdale	State: CA	Zip Code: 93656		
Contact Person: Vincent Romero		Telephone Number: (559) 867-3838	Federal Tax ID: 94-6021486	

C. Facility Operator (The agency or business, not the person):

Name: same as owner			Operator Type (Check One) 1. ☐ Individual 2. ☐ Corporation 3. ☐ Governmental Agency 4. ☐ Partnership 5. ☐ Other: _____	
Address:				
City:	State:	Zip Code:		
Contact Person:		Telephone Number:		

D. Owner of the Land:

Name: same as owner			Owner Type (Check One) 1. ☐ Individual 2. ☐ Corporation 3. ☐ Governmental Agency 4. ☐ Partnership 5. ☐ Other: _____	
Address:				
City:	State:	Zip Code:		
Contact Person:		Telephone Number:		

E. Address Where Legal Notice May Be Served:

Address: same as owner		
City:	State:	Zip Code:
Contact Person:		Telephone Number:

F. Billing Address:

Address: same as owner		
City:	State:	Zip Code:
Contact Person:		Telephone Number:

CALIFORNIA ENVIRONMENTAL
PROTECTION AGENCY



State of California
Regional Water Quality Control Board

**APPLICATION/REPORT OF WASTE DISCHARGE
GENERAL INFORMATION FORM FOR
WASTE DISCHARGE REQUIREMENTS OR NPDES PERMIT**



II. TYPE OF DISCHARGE

Check Type of Discharge(s) Described in this Application (A or B):

☒ **A. WASTE DISCHARGE TO LAND**

☐ **B. WASTE DISCHARGE TO SURFACE WATER**

Check all that apply:

- ☒ Domestic/Municipal Wastewater Treatment and Disposal
☐ Cooling Water
☐ Mining
☐ Waste Pile
☐ Wastewater Reclamation
☐ Other, please describe: _____

- ☐ Animal Waste Solids
☐ Land Treatment Unit
☐ Dredge Material Disposal
☐ Surface Impoundment
☐ Industrial Process Wastewater

- ☐ Animal or Aquacultural Wastewater
☐ Biosolids/Residual
☐ Hazardous Waste (see instructions)
☐ Landfill (see instructions)
☐ Storm Water

III. LOCATION OF THE FACILITY

Describe the physical location of the facility.

1. Assessor's Parcel Number(s)
 Facility: 053-090-1100
 Discharge Point: 053-090-1100

2. Latitude
 Facility: 36deg 27' 25"N
 Discharge Point: 36deg 27' 25"N

3. Longitude
 Facility: 119deg 52'00"
 Discharge Point: 119deg 52'00"

IV. REASON FOR FILING

- ☐ New Discharge or Facility
☐ Changes in Ownership/Operator (see instructions)
☒ Change in Design or Operation
☐ Waste Discharge Requirements Update or NPDES Permit Reissuance
☒ Change in Quantity/Type of Discharge
☐ Other: _____

V. CALIFORNIA ENVIRONMENTAL QUALITY ACT (CEQA)

Name of Lead Agency: Riverdale Public Utility District

Has a public agency determined that the proposed project is exempt from CEQA? ☐ Yes ☒ No

If Yes, state the basis for the exemption and the name of the agency supplying the exemption on the line below.

Basis for Exemption/Agency: _____

Has a "Notice of Determination" been filed under CEQA? ☐ Yes ☒ No

If Yes, enclose a copy of the CEQA document, Environmental Impact Report, or Negative Declaration. If no, identify the expected type of CEQA document and expected date of completion.

Expected CEQA Documents: Initial Study/Mitigated Negative Declaration

☐ EIR ☒ Negative Declaration

Expected CEQA Completion Date: April 2017

CALIFORNIA ENVIRONMENTAL
PROTECTION AGENCY



State of California
Regional Water Quality Control Board

**APPLICATION/REPORT OF WASTE DISCHARGE
GENERAL INFORMATION FORM FOR
WASTE DISCHARGE REQUIREMENTS OR NPDES PERMIT**



VI. OTHER REQUIRED INFORMATION

Please provide a COMPLETE characterization of your discharge. A complete characterization includes, but is not limited to, design and actual flows, a list of constituents and the discharge concentration of each constituent, a list of other appropriate waste discharge characteristics, a description and schematic drawing of all treatment processes, a description of any Best Management Practices (BMPs) used, and a description of disposal methods.

Also include a site map showing the location of the facility and, if you are submitting this application for an NPDES permit, identify the surface water to which you propose to discharge. Please try to limit your maps to a scale of 1:24,000 (7.5' USGS Quadrangle) or a street map, if more appropriate.

VII. OTHER

Attach additional sheets to explain any responses which need clarification. List attachments with titles and dates below:

Riverdale Public Utility District Report of Waste Discharge for WWTP Improvement Project, January 20, 2017

You will be notified by a representative of the RWQCB within 30 days of receipt of your application. The notice will state if your application is complete or if there is additional information you must submit to complete your Application/Report of Waste Discharge, pursuant to Division 7, Section 13260 of the California Water Code.

VIII. CERTIFICATION

"I certify under penalty of law that this document, including all attachments and supplemental information, were prepared under my direction and supervision in accordance with a system designed to assure that qualified personnel properly gathered and evaluated the information submitted. Based on my inquiry of the person or persons who manage the system, or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment."

Print Name: Vincent Romero

Title: District Superintendent

Signature: V Romero

Date: 1-23-2017

FOR OFFICE USE ONLY

Date Form 200 Received:	Letter to Discharger:	Fee Amount Received:	Check #:
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Appendix B
WDR Order No. 85-252

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 85-252

WASTE DISCHARGE REQUIREMENTS
FOR
RIVERDALE PUBLIC UTILITIES DISTRICT
FRESNO COUNTY

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Board) finds that:

1. Riverdale Public Utilities District (hereafter Discharger) submitted a Report of Waste Discharge, dated 30 December 1983.
2. The Board, on 27 April 1979, adopted Order No. 79-101 which prescribed requirements for a discharge from a trickling filter to percolation/evaporation ponds.
3. Present waste discharge requirements are neither adequate nor consistent with plans and policies of the Board.
4. The Discharger currently discharges an average of 140,000 gallons per day (0.6 l/sec) from stabilization ponds to pasture land regulated by Wastewater Reclamation Order No. 79-28. Design flow for the facility is 250,000 gpd (10.9 l/sec).
5. The treatment and disposal facilities are in Section 13, T17S, R19E, MDB&M.
6. The beneficial uses of the ground water are municipal, industrial, and agricultural supply.
7. The Board, on 25 July 1975, adopted a Water Quality Control Plan for the Tulare Lake Basin (5D) which contains water quality objectives. These requirements are consistent with that Plan.
8. The action to adopt waste discharge requirements for this existing facility is exempt from the provisions of the California Environmental Quality Act, in accordance with Section 15301, Title 14, California Administrative Code.
9. The Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge.
10. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Order No. 79-101 be rescinded and Riverdale Public Utilities District, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following:

WASTE DISCHARGE REQUIREMENTS
RIVERDALE PUBLIC UTILITIES DISTRICT
FRESNO COUNTY

-2-

A. Discharge Prohibitions:

1. The direct discharge of wastes to surface waters or surface water drainage courses is prohibited.
2. The by-pass or overflow of untreated or partially treated waste is prohibited.

B. Discharge Specifications:

1. Neither the treatment nor the discharge shall cause a pollution or nuisance as defined by the California Water Code, Section 13050.
2. The discharge shall not cause degradation of any water supply.
3. The discharge shall remain within the designated disposal area at all times.
4. The 30-day average daily dry weather discharge flow shall not exceed 0.250 million gallons (946 m³).
5. Collected screenings, sludges, and other solids removed from liquid wastes shall be disposed of in a manner approved by the Executive Officer.
6. Reclaimed wastewater shall meet the criteria contained in Title 22, Division 4, California Administrative Code (Section 60301, et seq.).
7. The discharge from the treatment facilities shall not contain constituents in excess of the following limits:

<u>Constituents</u>	<u>Units</u>	<u>30-Day Mean</u>	<u>Maximum</u>
BOD	mg/l	40	80
Settleable Matter	mg/l	0.5	1.0

8. The dissolved oxygen content of holding ponds shall not be less than 1.0 mg/l for 16 hours in any 24-hour period.
9. The Discharger shall preclude public access to treatment and disposal facilities through a satisfactory combination of fencing and posting or an acceptable alternative.
10. The maximum electrical conductivity (EC) of the discharge shall not exceed 1100 micromhos.

WASTE DISCHARGE REQUIREMENTS
RIVERDALE PUBLIC UTILITIES DISTRICT
FRESNO COUNTY

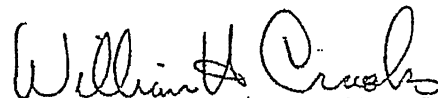
-3-

11. Pasture shall not be irrigated while cattle are present.

C. Provisions:

1. The Discharger may be required to submit technical reports as directed by the Executive Officer.
2. The Discharger shall comply with the attached Monitoring and Reporting Program No. 85-252.
3. The Discharger shall comply with the Standard Provisions and Reporting Requirements, dated 1 September 1985, which are a part of this Order.
4. The Discharger shall report promptly to the Board any material change or proposed change in the character, location, or volume of the discharge.
5. In the event of any change in control or ownership of land or waste discharge facilities presently owned or controlled by the Discharger, the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be forwarded to this office.
6. The Board will review this Order periodically and may revise requirements when necessary.

I, WILLIAM H. CROOKS, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 27 September 1985.



WILLIAM H. CROOKS, Executive Officer

LDT:bro:5/20/85

Attachments

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

MONITORING AND REPORTING PROGRAM NO. 85-252

FOR
RIVERDALE PUBLIC UTILITIES DISTRICT
FRESNO COUNTY

EFFLUENT MONITORING

Effluent samples shall be collected downstream from the last connection through which wastes can be admitted to the disposal ponds. Effluent samples should be representative of the volume and nature of the discharge. The following shall constitute the effluent monitoring program:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
Total Daily Flow	mgd, l/s	Continuous	Daily
20°C BOD ₅	mg/l	Grab	Monthly
Settleable Matter	ml/l	Grab	2/month
Dissolved Oxygen ^{1/}	mg/l	Grab	Weekly
Specific Electrical Conductance	umhos/cm	Grab	Weekly

^{1/} Sample to be collected from a location opposite each pond inlet between the hours of 0800 and 0900.

If the discharge is intermittent rather than continuous, then on the first day of each such intermittent discharge, the Discharger shall monitor and record data for all of the constituents listed above, after which the frequencies of analysis given in the schedule shall apply for the duration of each such intermittent discharge. In no event shall the Discharger be required to monitor and record data more often than twice the frequencies listed in the schedule.

REPORTING

In reporting the monitoring data, the Discharger shall arrange the data in tabular form so that the date, the constituents, and the concentrations are readily discernible. The data shall be summarized in such a manner to illustrate clearly the compliance with waste discharge requirements.

MONITORING AND REPORTING PROGRAM
RIVERDALE PUBLIC UTILITIES DISTRICT
FRESNO COUNTY

-2-

Monthly monitoring reports shall be submitted to the Regional Board by the 15th day of the following month.

The results of any monitoring done more frequently than required at the locations specified in the Monitoring and Reporting Program shall be reported to the Board.

Ordered by

William H. Crooks
WILLIAM H. CROOKS, Executive Officer

27 September 1985

(Date)

LDT:bro

5/20/85

INFORMATION SHEET

RIVERDALE PUBLIC UTILITIES DISTRICT FRESNO COUNTY

Riverdale Public Utilities District owns and operates wastewater treatment and disposal facilities in Section 13, T17S, R19E, MDB&M. The District serves the City of Riverdale and surrounding area.

The treatment facility itself consists of a comminutor, an aeration pond, and six stabilization ponds operated in series. Two mechanical aerators operate in the aeration pond. Stabilization ponds are rip-rapped to inhibit wind erosion. The design capacity of the plant is 250,000 gpd (10.91/sec). Present daily flows average 140,000 gpd (0.6 l/sec).

Disposal is accomplished on 36 acres (14.6 ha) of neighboring pasture. Disposal is regulated by Wastewater Reclamation Order No. 79-28, on land owned by Mr. Michael Harmon.

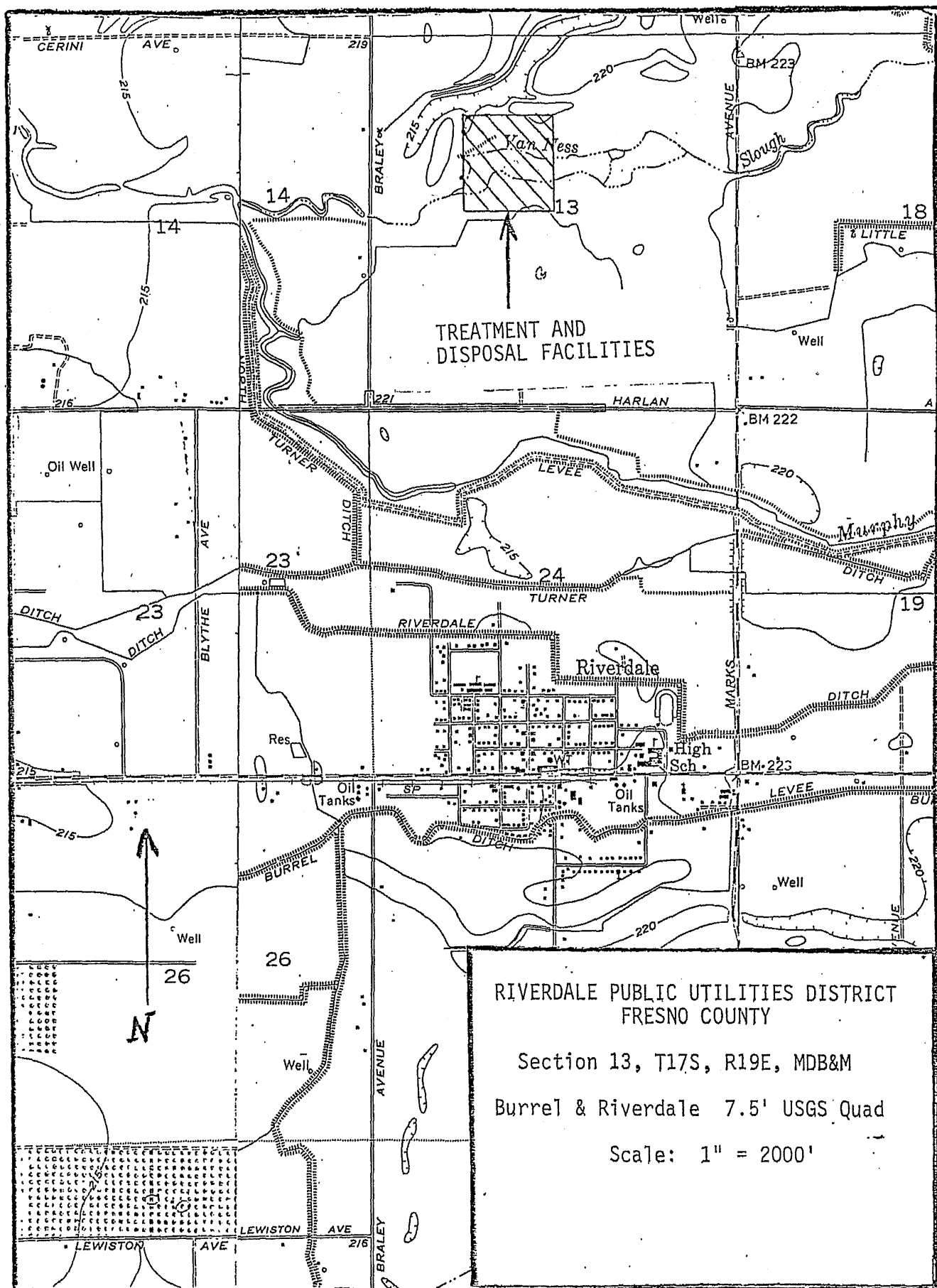
The Department of Water Resources (DWR) records an annual average precipitation of 7.59 inches (19.3 cm) at Riverdale over the 54-year period of record ending in 1975. For the 5-year period of record ending in 1985, DWR reports an annual average evaporation of 59.76 inches (151.8 cm). Daily temperatures in July average 82.6°F at Five Points, and 45.9°F in December.

Depth-to-water levels at wells in Section 14 average 85 feet (25.5 m). Ground water quality data of samples from Section 15 record conductivities averaging 613 umhos/cm.

The action to adopt revised waste discharge requirements for this existing facility is exempt from the provision of the California Environmental Quality Act, in accordance with Section 15301, Title 14, Chapter 3, California Administrative Code.

LDT:bro

5/20/85



Appendix C

Water Balance Calculations

**Riverdale Public Utility District
Wastewater Treatment Plant Disposal
Normal Year Rainfall - 0.275 MGD**

DATA:

Month	Number of Days per Month	Working Days per Month	Normal Year Rainfall (in/month)	Evaporation (in/month)	ET c (in/month)
January	31	31	2.02	0.91	0.79
February	28	28	1.83	2.65	2.07
March	31	31	1.84	5.01	3.89
April	30	30	0.36	6.95	4.74
May	31	31	0.13	9.15	6.15
June	30	30	0.13	9.38	6.46
July	31	31	0.01	9.98	6.61
August	31	31	0.01	8.45	5.49
September	30	30	0.15	8.34	4.32
October	31	31	0.31	4.34	1.82
November	30	30	1.12	1.31	0.80
December	31	31	1.65	1.24	1.49
Total	365	365	10.68	66.20	44.57

Effluent Production =	275,000	gpd
Treatment Pond (Lined)	1.80	ac-ft
Approx. Storage Volume =	1.1	ac-ft
Approx. Storage Volume =	2,411,000	gal
Pond Percolation Rate =	0.00	in/day
Existing Ponds 5 and 6	2.41	ac-ft
Approx. Storage Volume =	3,561,000	gal
Approx. Storage Volume =	5,170,000	gal
Pond Percolation Rate =	0.33	in/day
Additional Ponds	21.7	ac-ft
Approx. Storage Volume =	457,000	gal
Approx. Storage Volume =	29,497,000	gal
Pond Percolation Rate =	0.33	in/day

STORAGE POND CALCULATIONS:

Month	Effluent Produced (gall/month)	Effluent to Alafia (gall/month)	Effluent to Ponds (gall/month)	Surface Rainfall (gall/month)	Surface Evaporation (gall/month)	Pond Percolation (gall/month)	Monthly Available (gall/month)	Cumulative Available (gall/month)
January	8,325,000	0	8,325,000	1,421,207	1,864,454	6,697,477	2,606,725	8,048,685
February	7,700,000	0	7,700,000	1,281,529	1,584,454	6,049,344	1,073,741	9,121,826
March	8,325,000	0	8,325,000	1,294,564	3,528,633	6,697,477	-404,546	8,717,280
April	8,325,000	0	8,325,000	1,244,675	4,889,795	6,697,477	3,437,006	12,154,286
May	8,325,000	0	8,325,000	353,284	6,437,643	6,697,477	-4,356,836	7,797,450
June	8,325,000	0	8,325,000	105,535	6,666,303	6,481,430	-4,921,198	3,005,252
July	8,325,000	0	8,325,000	7,036	7,018,087	4,519,201	-3,005,252	0
August	8,325,000	0	8,325,000	105,535	5,945,146	2,586,890	0	0
September	8,325,000	0	8,325,000	358,819	4,740,287	3,615,248	0	0
October	8,325,000	0	8,325,000	787,996	923,432	6,481,430	1,653,134	3,323,615
November	8,325,000	0	8,325,000	1,160,887	870,665	6,697,477	2,117,745	5,441,360
December	8,325,000	0	8,325,000	7,514,103	44,020,311	63,868,792	0	Start at 0 Stored October 1st
Total	100,375,000	0	100,375,000	23	135	196		

Alafia Area (available) = 0 ac-ft
 Alafia Area (needed) = 3.00 ac-ft
 Alafia Rootzone AVHC = 3.00 inch

RECLAMATION AREA:

Month	Effluent Applied (gall/month)	Effluent Applied (in)	Effective Rainfall (in)	Fresh Irrigation (in)	Gross Crop Need (in)	Soil Moisture Start (in)	Soil Moisture End (in)	Percolation & Leaching (in)
January	0	#DIV/0!	1.36	0.00	2.96	#DIV/0!	#DIV/0!	#DIV/0!
February	0	#DIV/0!	1.18	0.00	5.56	#DIV/0!	#DIV/0!	#DIV/0!
March	0	#DIV/0!	0.47	0.00	6.77	#DIV/0!	#DIV/0!	#DIV/0!
April	0	#DIV/0!	0.00	0.00	8.79	#DIV/0!	#DIV/0!	#DIV/0!
May	0	#DIV/0!	0.00	0.00	9.23	#DIV/0!	#DIV/0!	#DIV/0!
June	0	#DIV/0!	0.00	0.00	9.44	#DIV/0!	#DIV/0!	#DIV/0!
July	0	#DIV/0!	0.00	0.00	7.84	#DIV/0!	#DIV/0!	#DIV/0!
August	0	#DIV/0!	0.00	0.00	6.46	#DIV/0!	#DIV/0!	#DIV/0!
September	0	#DIV/0!	0.26	0.00	2.83	#DIV/0!	#DIV/0!	#DIV/0!
October	0	#DIV/0!	0.51	0.00	1.14	#DIV/0!	#DIV/0!	#DIV/0!
November	0	#DIV/0!	1.01	0.00	1.53	#DIV/0!	#DIV/0!	#DIV/0!
December	0	#DIV/0!	4.79	0.00	63.68	#DIV/0!	#DIV/0!	#DIV/0!
Total	0	#DIV/0!	0.0	0.0	0.0	0.0	0.0	0.0

Percent of Total = #DIV/0!

- Assumptions
1. Rainfall Data per the Western Regional Climate Center.
 2. Evaporation data based on 125% of Evapotranspiration (Eto) based on DWR correspondence.
 3. Evapotranspiration per the CIP-Poly Irrigation Training & Research Center (ITRC).
 4. Percolation rates are a conservative estimate based on soil types in the area.

Total Pond Storage Available (gal) 33,600,000
 Cumulative Storage Needed (gal) 12,154,286
 Check OK

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 ENGINEERING GROUP

**Riverdale Public Utility District
Wastewater Treatment Plant Disposal
Wet Year Rainfall - 0.275 MGD**

DATA:

Month	Number of Days per Month	Working Days per Month	Wet Year Rainfall (in/month)	Evaporation (in/month)	At-A-Rate (in/month)	Effluent Production (gpd)
January	31	31	3.14	0.49	0.41	275,000
February	28	28	3.70	1.01	0.88	1,800
March	31	31	4.53	3.45	2.97	1,800
April	30	30	2.76	5.15	4.49	1,800
May	31	31	0.01	5.10	4.47	1,800
June	30	30	0.00	7.89	5.89	1,800
July	31	31	0.09	8.75	5.74	1,800
August	31	31	1.03	5.98	3.90	1,800
September	30	30	2.51	1.31	1.06	1,800
October	31	31	1.75	1.28	1.10	1,800
November	30	30	1.75	1.28	1.10	1,800
December	31	31	1.75	1.28	1.10	1,800
Total	365	365	21.61	54.11	39.00	39,000

At-A-Rate Area (Available) = 0 ac-ft
At-A-Rate Area = 0 ac-ft
At-A-Rate Area (Available) = 0 ac-ft
At-A-Rate Area = 0 ac-ft

RECLAMATION AREA:

Month	Effluent Applied (in/month)	Effluent Applied (in)	Effective Rainfall (in)	Evaporation (in)	Net (in)	Gross Crop (in)	Soil Moisture Start (in)	Soil Moisture End (in)	Percolation & Evaporation (in)
January	0	0	4.29	0.00	0.59	0.59	0.59	0.59	0.59
February	0	0	2.94	0.00	1.26	1.26	0.59	0.59	0.59
March	0	0	2.72	0.00	4.24	4.24	0.59	0.59	0.59
April	0	0	0.00	0.00	6.41	6.41	0.59	0.59	0.59
May	0	0	0.00	0.00	6.39	6.39	0.59	0.59	0.59
June	0	0	0.00	0.00	8.41	8.41	0.59	0.59	0.59
July	0	0	0.00	0.00	8.84	8.84	0.59	0.59	0.59
August	0	0	0.00	0.00	8.20	8.20	0.59	0.59	0.59
September	0	0	0.00	0.00	5.57	5.57	0.59	0.59	0.59
October	0	0	0.00	0.00	2.71	2.71	0.59	0.59	0.59
November	0	0	1.82	0.00	1.51	1.51	0.59	0.59	0.59
December	0	0	1.11	0.00	1.57	1.57	0.59	0.59	0.59
Total	0	0	12.88	0.00	55.70	55.70	0.59	0.59	0.59

Percent of Total =

STORAGE POND CALCULATIONS:

Month	Produced to At-A-Rate (gall/month)	Effluent to Ponds (gall/month)	Surface Rainfall (gall/month)	Surface Evaporation (gall/month)	Pond Percolation (gall/month)	Monthly Available (gall/month)	Cumulative Available (gall/month)
January	8,525,000	0	8,525,000	2,693,200	342,989	6,697,477	3,541,584
February	7,700,000	0	8,525,000	3,181,161	2,421,308	6,697,477	10,239,061
March	8,525,000	0	8,525,000	1,941,847	3,023,373	6,697,477	16,936,538
April	8,525,000	0	8,525,000	7,006	3,588,195	6,697,477	23,634,015
May	8,525,000	0	8,525,000	0	3,549,389	6,697,477	30,331,492
June	8,525,000	0	8,525,000	0	6,587,152	6,697,477	37,028,969
July	8,525,000	0	8,525,000	63,321	4,203,816	6,697,477	43,726,446
August	8,525,000	0	8,525,000	724,675	4,203,816	6,697,477	50,423,923
September	8,525,000	0	8,525,000	63,321	3,660,519	5,527,802	55,951,725
October	8,525,000	0	8,525,000	1,765,955	923,432	6,697,477	62,649,152
November	8,525,000	0	8,525,000	1,231,243	897,049	6,697,477	69,346,629
December	8,525,000	0	8,525,000	15,204,096	38,071,800	77,507,296	146,853,925
Total	100,375,000	0	100,375,000	47	117	28	146,853,925

Start of Stored October 1st

Assumptions:

1. Rainfall Data per the Western Regional Climate Center.
2. Evaporation data based on 12.5% of Evaporation (Ea) based on DWR correspondence.
3. Evaporation rate per the California Irrigation Training & Research Center (ITIRC).
4. Percolation rates are a conservative estimate based on soil types in the area.

Total Pond Storage Available (gal) 33,800,000
Cumulative Storage Needed (gal) 16,089,605
Check OK

Note: Sufficient storage is available cumulative volume after wet year could be stored over and evaporated and percolated during a normal year.



**Riverdale Public Utility District
Wastewater Treatment Plant Disposal
Wet Year Rainfall - 0.325 MGD**

DATA:

Month	Number of Days per Month	Working Days per Month	Wet Year Rainfall (in/month)	ET c Evaporation (in/month)	Alfalfa
January	31	31	5.14	0.49	0.41
February	28	28	3.70	1.01	0.88
March	31	31	4.53	3.45	2.97
April	30	30	2.76	5.15	4.49
May	31	31	0.01	5.10	4.47
June	30	30	0.00	7.89	5.89
July	31	31	0.00	9.36	6.19
August	31	31	0.09	8.75	5.74
September	30	30	1.03	5.98	3.90
October	31	31	0.09	4.35	1.90
November	30	30	2.51	1.31	1.06
December	31	31	1.75	1.28	1.10
Total	365	365	21.61	54.11	39.00

STORAGE POND CALCULATIONS:

Month	Effluent Produced (gal/month)	Effluent to Alfalfa (gal/month)	Effluent to Ponds (gal/month)	Surface Rainfall (gal/month)	Surface Evaporation (gal/month)	Pond Percolation (gal/month)	Monthly Available (gal/month)	Cumulative Available (gal/month)
January	10,075,000	0	10,075,000	4,288,373	403,883	7,975,304	5,394,186	11,471,474
February	9,100,000	0	9,100,000	3,065,366	838,833	7,203,304	4,123,033	15,594,507
March	10,075,000	0	10,075,000	3,793,002	2,835,247	7,975,304	2,994,451	18,588,958
April	9,750,000	0	9,750,000	2,886,597	4,266,638	7,718,036	51,903	18,640,861
May	10,075,000	0	10,075,000	8,385	4,225,234	7,975,304	2,117,253	16,758,114
June	9,750,000	0	9,750,000	0	6,534,615	7,718,036	4,502,651	12,255,463
July	10,075,000	0	10,075,000	74,563	7,249,177	7,975,304	5,074,918	17,330,381
August	10,075,000	0	10,075,000	853,332	4,950,152	6,942,296	1,289,116	18,619,497
September	9,750,000	0	9,750,000	0	3,603,876	6,545,687	0	18,619,497
October	10,075,000	0	10,075,000	2,079,478	1,087,376	7,718,036	3,024,066	21,643,563
November	9,750,000	0	9,750,000	1,499,835	1,056,309	7,975,304	2,493,222	24,136,785
December	10,075,000	0	10,075,000	17,903,394	44,830,979	91,697,415	0	24,136,785
Total	118,625,000	0	118,625,000	55	138	281		

Alfalfa Area (Available) = 0 acres
Alfalfa Area = 0 acres
Alfalfa Rootzone AWHC = 3.00 inch

RECLAMATION AREA:

Month	Alfalfa Effluent Applied (gal/month)	Effluent Applied (in)	Effective Rainfall (in)	Fresh Irrigation (in)	Gross Crop Need (in)	Soil Moisture Start (in)	Soil Moisture End (in)	Irrigation Application Efficiency = 70%	Leaching* Percolation & Moisture (in)
January	0	#DIV/0!	4.29	0.00	0.59	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
February	0	#DIV/0!	2.94	0.00	1.26	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
March	0	#DIV/0!	2.72	0.00	4.24	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
April	0	#DIV/0!	0.00	0.00	6.41	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
May	0	#DIV/0!	0.00	0.00	6.39	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
June	0	#DIV/0!	0.00	0.00	8.41	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
July	0	#DIV/0!	0.00	0.00	8.84	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
August	0	#DIV/0!	0.00	0.00	8.20	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
September	0	#DIV/0!	0.00	0.00	5.57	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
October	0	#DIV/0!	0.00	0.00	2.71	3.00	#DIV/0!	#DIV/0!	#DIV/0!
November	0	#DIV/0!	1.83	0.00	1.51	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
December	0	#DIV/0!	1.11	0.00	1.57	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
Total	0	#DIV/0!	12.88	0.00	55.70	0.0	0.0	0.0	0.0

Percent of Total = 0.0 ac-ft #DIV/0! #DIV/0! #DIV/0! #DIV/0! #DIV/0! #DIV/0! #DIV/0! #DIV/0! #DIV/0! #DIV/0!

* Leaching Ratio (LR) = $EC_e / (EC_e - EC_r)$ 20%

LR = % of total applied water to maintain the average root zone salinity at ECe while irrigating with irrigation water quality ECe.

Total Pond Storage Available (gal) 39,800,000
Cumulative Storage Needed (gal) 18,440,861
Check OK

- Assumptions**
1. Rainfall Data per the Western Regional Climate Center.
 2. Evaporation data based on 125% of Evapotranspiration (Eto) based on DWR correspondence.
 3. Evapotranspiration per the CalPoly Irrigation Training & Research Center (ITRC).
 4. Percolation rates are a conservative estimate based on soil types in the area.



Appendix D Reclamation Letter

Riverdale Public Utility District

Post Office Box 248
20896 Malsbary
Riverdale, California, 93656

Phone (559) 867-3838
FAX (559) 867-3182

DIRECTORS:

Chad Swearingen, President
Roger Harrison, Director
Tommy Fritz, Director
Mayra Chavez, Director
Tristan Mauger, Director

November 29, 2016

Mark & Megan McKean
P.O. Box 445
Riverdale, CA 93656

RE: Riverdale Public Utility District (Riverdale PUD) Potential Recycle of Treated Effluent

The Riverdale PUD wastewater treatment plant (WWTP) near your property serves the community of Riverdale and currently treats an average of approximately 210,000 gallons of wastewater per day. Riverdale PUD is currently planning improvements to the WWTP. The wastewater effluent produced by the proposed treatment facilities is classified as undisinfected secondary effluent, and must achieve effluent characteristics no greater than 30 mg/L biochemical oxygen demand (BOD) and no greater than 30 mg/L total suspended solids (TSS). Due to high salinity levels in the Riverdale PUD supply water, the WWTP effluent has electrical conductivity (EC) levels of about 1,540 µmhos/cm, on average.

The State of California Regional Water Quality Control Board (RWQCB) encourages reuse (or "reclamation") of treated effluent for beneficial uses such as irrigation for agriculture use. Riverdale PUD would be interested in discussing the possibility of entering into an agreement for the purpose of reclamation of any portion of the treated effluent. Please note that if reclamation is desired, Riverdale PUD would need to prepare and submit an application to the RWQCB and the State Water Resources Control Board, Division of Drinking Water. Specific monitoring and reporting requirements for the Riverdale PUD and property owner where reclamation occurs would then be defined by the RWQCB. The establishment of a reclamation site would require conformance with the State Water Resources Control Board Order WQ 2014-0090-DWQ, General Waste Discharge Requirements for Recycled Water Use, Adopted June 3, 2014.

Pursuant to the California Code of Regulations, Article 3, Section 60304, potential surface irrigation uses for the effluent from the Riverdale PUD WWTP are as follows:

- (1) Orchard where the recycled water does not come into contact with the edible portion of the crop,

- (2) Vineyards where the recycled water does not come into contact with the edible portion of the crop,
- (3) Non food-bearing trees (Christmas tree farms are included in this category provided no irrigation with recycled water occurs for a period of 14 days prior to harvesting or allowing access by the general public),
- (4) Fodder and fiber crops and pasture for animals not producing milk for human consumption,
- (5) Seed crops not eaten by humans,
- (6) Food crops that must undergo commercial pathogen-destroying processing before being consumed by humans, and
- (7) Ornamental nursery stock and sod farms provided no irrigation with recycled water occurs for a period of 14 days prior to harvesting, retail sale, or allowing access by the general public.

A crop location map is enclosed, which shows alfalfa and grains/hay/forage crops in the area that could utilize the undisinfected secondary effluent from the Riverdale PUD WWTP for irrigation. If reclamation on your existing crop land is not desired, Riverdale PUD also has up to 33 acres of land at the WWTP site that could be leased to a farmer to grow crops onsite with reuse of treated effluent.

Please contact Maija Madec at (559) 326-1100 or mmadec@ppeng.com if you are interested in exploring the potential of using treated effluent from the Riverdale PUD WWTP for reclamation.

Respectfully,

Vincent Romero
Riverdale Public Utility District

Enclosures: Crop Location Map

cc: Maija Madec, Provost & Pritchard Consulting Group

Appendix E

Antidegradation Analysis

This Appendix is Bound Separately

Appendix F

Preliminary Engineering Report

This Appendix is Bound Separately